

5G RAN

Hyper Cell Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
None	None	TDD	None

- Editorial changes

Added the MML configurations for TRP overshoot coverage handover optimization. For details, see [4.4.1.2 Using MML Commands](#).

Revised the name of the spectral efficiency estimation function. For details, see [4.3.2.3 Function Impacts](#).

Revised the principles of inter-TRP coordinated scheduling. For details, see [4.1.2 Inter-TRP Coordinated Scheduling](#).

Added the impact relationship between uplink basic preallocation and Hyper Cell in FSD transmission mode. For details, see [4.3.2.3 Function Impacts](#).

Added the impact relationship between uplink smart preallocation and Hyper Cell in FSD transmission mode. For details, see [4.3.2.3 Function Impacts](#).

Revised descriptions in this document.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 06 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for multi-TRP multi-TRS optimization for Hyper Cell. For details, see: • 4.1.2 Inter-TRP Coordinated Scheduling • 4.3.2.1 Prerequisite Functions • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Modified parameter: Added the MULTI_TRP_MULTITRIS_OPTION option to the NRDUCellCsirs.CsirsExtSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Revised the constraints affecting cell activation due to license insufficiency. For details, see 4.3.1 Licenses.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the mutually exclusive relationship between Hyper Cell and downlink experience satisfaction guarantee. For details, see 4.3.2.2 Mutually Exclusive Functions.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added support for inter-TRP F4 HARQ resource sharing. For details, see: • 4.1.4 Other Performance Optimization Functions • 4.2.2 Impacts • 4.3.2.3 Function Impacts • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Modified parameter: Added the INTER_TRP_F4_HARQ_SHARE_SW option to the NRDUCellPucch.PucchAlgoExtSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for FSD transmission mode by Hyper Cell. For details, see: • 4.1 Principles • 4.1.2 Inter-TRP Coordinated Scheduling • 4.2.1 Benefits • 4.2.2 Impacts • 4.3.2.2 Mutually Exclusive Functions • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Modified parameter: Added the value FSD_MODE to the NRDUCellMultiTrp.DataTransMode parameter. Added parameters: • NRDUCellMultiTrp.UlIntrfrTrpSrsRsrpDiffThld • NRDUCellMultiTrp.DlIntrfrTrpSrsRsrpDiffThld	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Changed the reserved parameter for adaptive CSI-RS beam sweeping for CA UEs to a formal parameter. For details, see: • 4.1.4 Other Performance Optimization Functions • 4.2.1 Benefits • 4.2.2 Impacts • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	The CA_CSIRS_BM_SWEEP_ADPT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter replaces RSVDSWPARAM1_BIT18 of the NRDUCellRsvdExt03.RsvdSwParam1 parameter. • Modified parameter: Added the CA_CSIRS_BM_SWEEP_ADPT_SW option to the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter. • Added RSVDSWPARAM1_BIT18 of the NRDUCellRsvdExt03.RsvdSwParam1 parameter to the disuse list.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Removed the mutually exclusive relationships between Hyper Cell and Inter-TRP SSB Staggering and between Hyper Cell and High-speed Railway Superior Experience when the BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter is selected. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added a mutually exclusive relationship between Hyper Cell and UE-specific CSI-RS reconfiguration reduction. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the mutually exclusive relationship between Hyper Cell and PUSCH coordinated power control. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationships between Hyper Cell and adaptive baseband shutdown and between Hyper Cell and power saving first. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the mutually exclusive relationships between Hyper Cell and FWA uplink single-user 4-layer transmission, FWA device-pipe-based uplink dense codebooks, and FWA device-pipe-based uplink high power. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added support for TRP overshoot-coverage handover optimization. For details, see: 4.1.3 Inter-TRP Handover 4.4.1.1 Data Preparation	Modified parameter: Added the TRP_OVER_COV_HO_OPT_SW option to the NRDUCellMultiTrp.MultiTrpAlgoSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
Changed the baseband chip shutdown function to baseband shutdown for energy saving in the impacted functions. For details, see [4.3.2.3 Function Impacts](#).
Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-030205	Hyper Cell	4 Hyper Cell

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Hyper Cell	None	4 Hyper Cell

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Hyper Cell	Supported only in low frequency bands	4 Hyper Cell

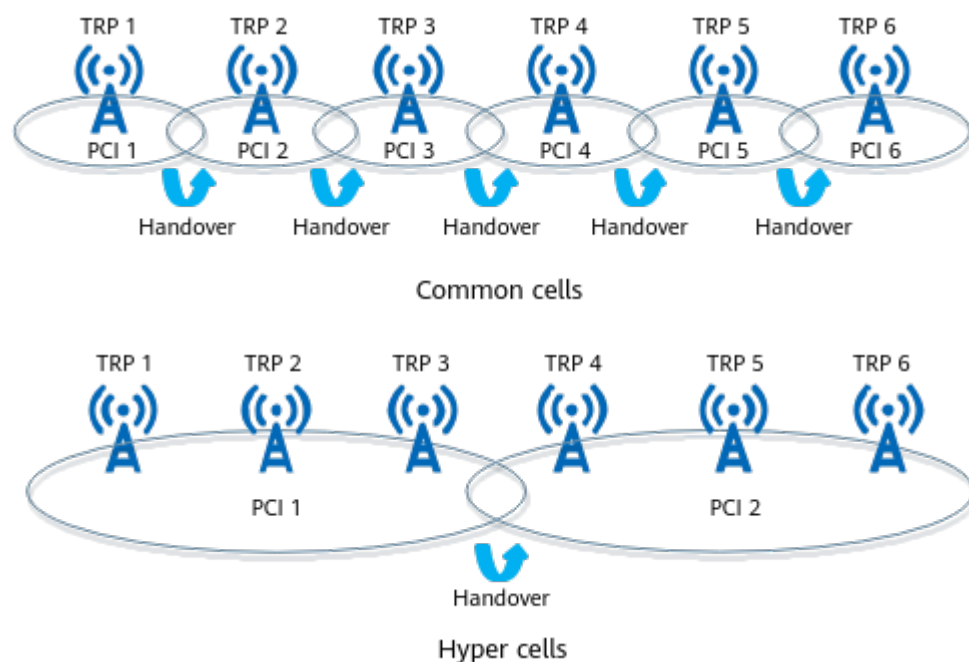
3 Overview

In linear coverage scenarios, such as subways or railways, when UEs move quickly along the coverage areas, a large number of handovers will occur, causing performance issues such as high service drop rate, difficult access, and low throughput. Hyper Cell is introduced to reduce the number of handovers and improve user experience in these scenarios.

Hyper Cell combines multiple transmission reception points (TRPs) that are working in the same frequency band and providing continuous coverage into one logical cell, whose physical cell identifier (PCI) is then used by all of its TRPs. No handover is required when UEs move in this logical cell. Hyper Cell decreases the number of handovers when UEs move fast between cells, thereby reducing performance deterioration caused by handovers.

Cells without TRP combination are referred to as common cells and cells corresponding to combined TRPs are referred to as hyper cells. [Figure 3-1](#) shows common cells and hyper cells in a typical continuous linear coverage scenario.

Figure 3-1 Example of common cells and hyper cells



For details about TRPs, see *Cell Management*.

4 Hyper Cell

4.1 Principles

After a hyper cell is formed, its frequency, bandwidth, PCI, and cell global identifier (CGI) are used by all of its TRPs. Inter-cell handovers are not required when UEs move between TRPs. [Table 4-1](#) describes key technologies for a hyper cell. The `NRDUCell.NrDuCellNetworkingMode` parameter must be set to `HYPER_CELL`, and the `NRDUCellMultiTrp.DataTransMode` parameter must be set to `BASIC_MODE` or `FSD_MODE`. The values `BASIC_MODE` and `FSD_MODE` represent the basic transmission mode and FSD transmission mode, respectively.

Table 4-1 Key technologies for a hyper cell

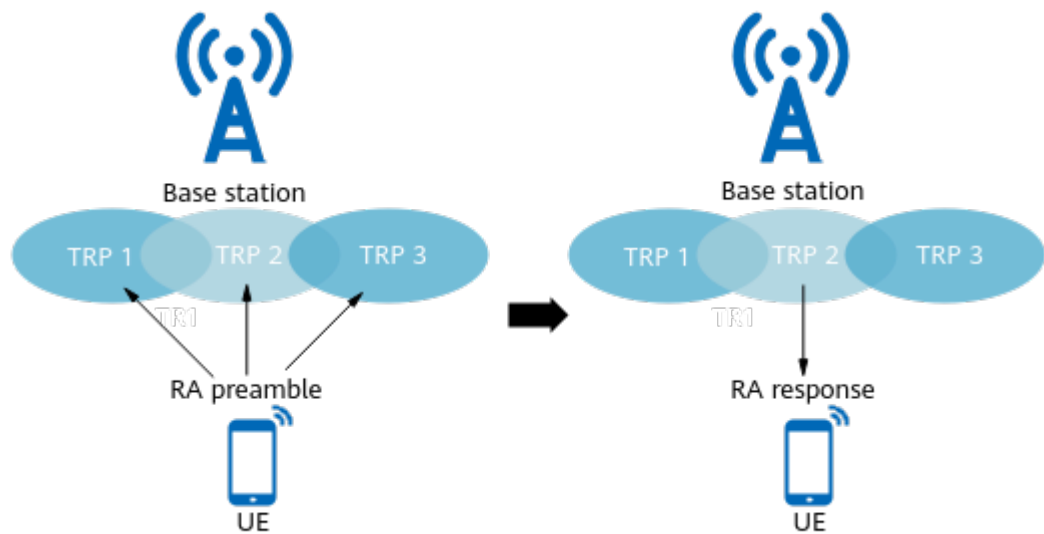
Key Technology	Description	Section
Serving TRP selection	During UE access, the gNodeB selects the TRP with the highest RSRP for random access (RA) preambles as the serving TRP for the UE.	4.1.1 Serving TRP Selection
Inter-TRP coordinated scheduling	The gNodeB schedules PUCCH, common PDCCH, and PRACH in a unified manner and allocates the same time-frequency resources to different TRPs. The serving TRP independently schedules uplink and downlink data channels (PUSCH and PDSCH) and the UE-dedicated PDCCH. Compared with the basic transmission mode, the FSD transmission mode considers user experience in overlapping areas, dynamically identifies UEs in overlapping areas and their uplink and downlink interfering TRPs, and performs real-time interference avoidance and scheduling. The gNodeB schedules uplink and downlink reference signals in a unified manner. Different types of reference signals are scheduled in different ways.	4.1.2 Inter-TRP Coordinated Scheduling

Key Technology	Description	Section
Inter-TRP handover	• SRS measurement-based fast handover When a UE moves between the TRPs for a hyper cell, the gNodeB measures the SRS RSRP and SRS SINR of the UE in each TRP to determine whether to perform an inter-TRP handover. • Inter-TRP handover in CA SCells When a UE for which asymmetric CA takes effect (that is, the UE uses different numbers of CCs in the uplink and downlink) moves between the TRPs for a hyper cell and it cannot send SRSs on secondary component carriers (SCCs), the gNodeB determines whether to perform an inter-TRP handover based on CSI-RS RSRP measured and reported by the UE in each TRP.	4.1.3 Inter-TRP Handover
Other performance optimization functions	Such functions include high-speed-specific interference randomization-based scheduling, TRS configuration optimization, PUSCH scrambling code optimization, inter-TRP RB resource sharing, CA flexible networking, CA enhancement in hyper cell co-coverage scenarios or combined cell co-coverage scenarios, CA flexible data split, inter-TRP F4 HARQ resource sharing, and adaptive CSI-RS beam sweeping for CA UEs.	4.1.4 Other Performance Optimization Functions

4.1.1 Serving TRP Selection

When a UE in a hyper cell accesses the network, the gNodeB measures the RA preamble RSRP of the UE in each TRP for the hyper cell, and then selects the TRP with the highest RSRP as the serving TRP for the UE, as shown in [Figure 4-1](#).

Figure 4-1 Serving TRP selection

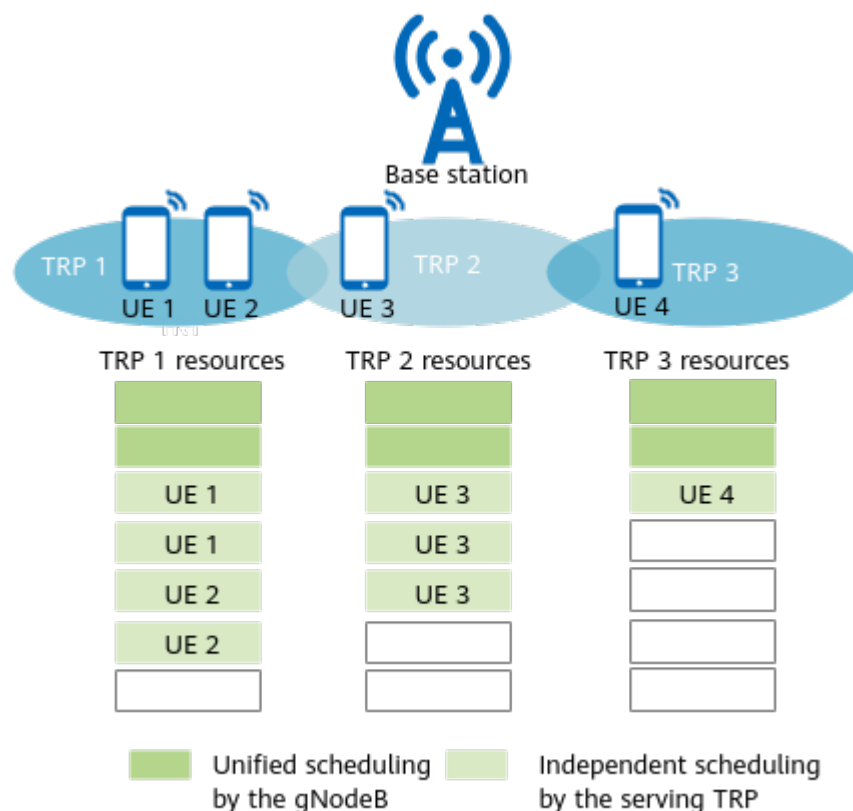


For details about RA preamble and RA response, see *Random Access Control*.

4.1.2 Inter-TRP Coordinated Scheduling

Coordinated scheduling between TRPs for a hyper cell involves unified scheduling by the gNodeB and independent scheduling by the serving TRP, as shown in [Figure 4-2](#).

Figure 4-2 Inter-TRP coordinated scheduling



In a hyper cell, the gNodeB schedules PUCCH, common PDCCH, and PRACH in a unified manner and allocates the same time-frequency resources to different TRPs.

Uplink and downlink data channels (PUSCH, PDSCH, and UE-dedicated PDCCH) of a hyper cell:

- Basic transmission mode: The serving TRP independently schedules uplink and downlink data channels (PUSCH and PDSCH) and the UE-dedicated PDCCH.
- FSD transmission mode:

In addition to the features of basic transmission mode, the FSD transmission mode considers user experience in overlapping areas. It dynamically identifies UEs in overlapping areas between TRPs and uplink and downlink interfering TRPs, and performs real-time interference avoidance on the identified interfering TRPs, thereby reducing inter-TRP interference on service and control channels. Compared with the basic transmission mode, the FSD transmission mode makes inter-TRP resource allocation more flexible and improves the performance of UEs in overlapping areas. The methods of identifying interfering TRPs are as follows:

- When the difference between the SRS RSRP of the serving TRP and that of another TRP is less than the value of **NRDUCellMultiTrp.UlIntrfTrpSrsRsrpDiffThld**, the gNodeB identifies the other TRP as an uplink interfering TRP for UEs in the overlapping area.
- When the difference between the SRS RSRP of the serving TRP and that of another TRP is less than the value of

NRDUCellMultiTrp.DlIntrfTrpSrsRsrpDiffThld, the gNodeB identifies the other TRP as the downlink interfering TRP for UEs in the overlapping area.

In a hyper cell, the uplink and downlink reference signals are scheduled in the ways described in [Table 4-2](#).

Table 4-2 Methods of scheduling uplink and downlink reference signals in a hyper cell

Reference Signal	Scheduling Method
Sounding reference signal (SRS)	The gNodeB schedules SRS resources in a unified manner and allocates the same time-frequency resources to different TRPs.
Synchronization signal and PBCH block (SSB)	• If the INTER_TRP_SSB_STAGGER_SWITCH option of the NRDUCellMultiTrp.MultiTrpAlgoSwitch parameter is selected: In TDD, the gNodeB staggers time-domain positions for SSB transmissions from different TRPs based on TRP ID mod 6 values. (A TRP ID is specified by the NRDUCellTrp.NrDuCellTrpId parameter.) • If this option is deselected: The gNodeB uses the same time-domain positions for SSB transmissions from different TRPs.

Reference Signal	Scheduling Method
Channel state information-reference signal (CSI-RS)	CSI-RS scheduling varies depending on the setting of the NRDUCellCsirs.InterTrpCsirsStaggerMode parameter. • If the parameter is set to NOT_STAGGER, one set of periodic CSI-RS resources is configured. • If the parameter is set to MOD2_STAGGER, two sets of periodic CSI-RS resources are configured for UEs that support the measurement using multiple sets of CSI-RS resources, and the gNodeB staggers time-domain positions for CSI-RS transmissions from different TRPs based on TRP ID mod 2 values. The following CSI-RS-related functions are provided as well: • CSI-RS interference measurement optimization: It is controlled by the INTRF_MEAS_OPT_SW option of the NRDUCellMultiTrp.MultiTrpCsiRsAlgoSwitch parameter. It is recommended that the function be enabled when the PRB usage of each TRP for a hyper cell exceeds 30% to optimize CSI-RS measurements in strong interference scenarios. • Inter-TRP handover in CA SCells: If CA UEs cannot obtain SRS resources on SCCs, inter-TRP handover in CA SCells can be enabled. The function takes effect based on TRP-specific resource allocation for CSI-RS beam sweeping and CSI-RS RSRP measurements. For details, see Inter-TRP Handover in CA SCells in 4.1.3 Inter-TRP Handover.
Tracking reference signal (TRS)	Different resources are allocated based on TRP ID mod 6 values. TRP scheduling varies depending on the setting of the NRDUCellCsirs.TrsAllocationStrategy parameter. • If the parameter is set to SINGLE_TRS, the gNodeB allocates one set of TRS resources in a hyper cell. • If the parameter is set to MULTI_TRS, the gNodeB allocates multiple sets of TRS resources in a hyper cell. It is recommended that this parameter be set to MULTI_TRS to improve the reliability of UEs moving between TRPs. It is recommended that the multi-TRP multi-TRS optimization (controlled by the MULTI_TRP_MULTI_TRS_OPT_SW option of the NRDUCellCsirs.CsiRsExtSwitch parameter) be enabled. This function allocates multiple sets of TRS resources based on the actual TRP deployment in a cell, reducing the TRS measurement capability overhead of UEs.

For details about channels and reference signals, such as PUCCH, common PDCCH, and SRS, see *Channel Management*.

For details about CA, see *Carrier Aggregation*.

4.1.3 Inter-TRP Handover

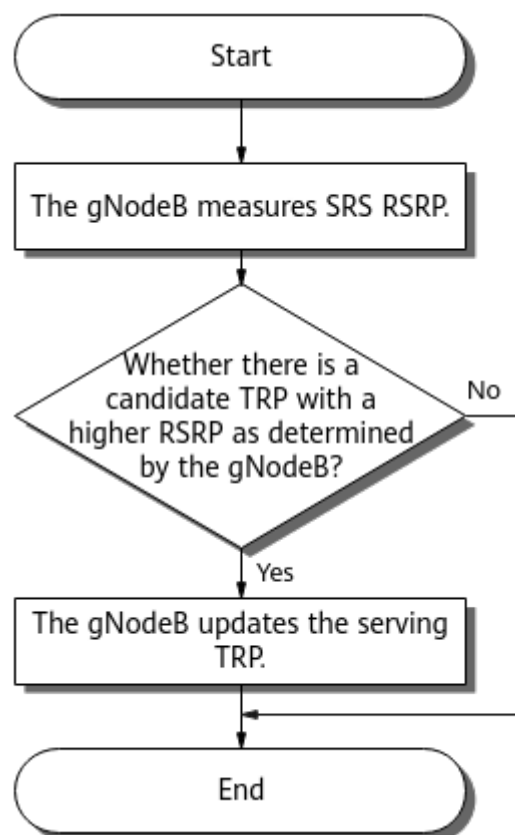
Inter-TRP handover involves the following key techniques:

- SRS measurement-based fast handover
When a UE moves between the TRPs for a hyper cell, the gNodeB measures the SRS RSRP and SRS SINR of the UE in each TRP to determine whether to perform an inter-TRP handover.
- Inter-TRP handover in CA SCells
When a UE for which asymmetric CA takes effect (that is, the UE uses different numbers of CCs in the uplink and downlink) moves between the TRPs for a hyper cell and it cannot send SRSs on secondary component carriers (SCCs), the gNodeB determines whether to perform an inter-TRP handover based on CSI-RS RSRP measured and reported by the UE in each TRP.

SRS Measurement-based Fast Handover

When a UE moves between the TRPs for a hyper cell, the gNodeB measures the SRS RSRP and SRS SINR of the UE in each TRP to determine whether to perform an inter-TRP handover. The specific process is shown in [Figure 4-3](#).

Figure 4-3 Process of an SRS measurement-based fast handover



1. The gNodeB measures SRS RSRP.
The gNodeB measures the SRS RSRP of a UE in each TRP.

2. The gNodeB determines whether there is a candidate TRP with a higher RSRP. The gNodeB uses the measurement result to determine whether there is a candidate TRP with a higher RSRP.
If the number of consecutive times the RSRP of a candidate TRP meets a certain evaluation condition reaches the value of the **NRDUCellMultiTrp.InterTrpHoRsrpEvalTimes** parameter, an inter-TRP handover is triggered. When a UE is handed over between TRPs for a hyper cell, the gNodeB does not need to send a handover command to the UE and the UE does not need to initiate an access procedure in the new serving TRP. The evaluation condition for triggering an inter-TRP handover is as follows: (RSRP of a candidate TRP – RSRP of the serving TRP) > (RSRP difference used to evaluate triggering of an inter-TRP handover + RSRP threshold offset for ping-pong handover prevention). Related parameters are described in [Table 4-3](#).

Table 4-3 Parameters related to SRS measurement-based fast handover

Parameter ID	Meaning	Description
NRDUCellMultiTrp.InterTrpHoRsrpEvalTimes	Specifies the number of evaluation times for an inter-TRP handover based on RSRP.	If the number of consecutive times a candidate TRP meets the evaluation condition reaches the value of this parameter, an inter-TRP handover is triggered. When signals fluctuate violently, you can set the parameter to a larger value to increase the number of RSRP evaluation times for an inter-TRP handover, avoiding ping-pong handovers between TRPs. When signals attenuate quickly, you can set the parameter to a smaller value to reduce the number of evaluation times, ensuring a prompt inter-TRP handover.
NRDUCellMultiTrp.HyperCellTrpHoRsrpPolicy	Specifies the RSRP policy for inter-TRP handovers.	- The value UL_RSRP indicates that the evaluation is based on the SRS RSRP. - The value DL_EQ_RSRP indicates that the evaluation is based on the SRS RSRP and the total power of the channels in the TRP.

Parameter ID	Meaning	Description
NRDUCellMultiTrp.TrpHoDeltaRsrp	Specifies the RSRP difference for an inter-TRP handover.	A larger value of this parameter results in a lower probability of triggering inter-TRP handovers and a lower probability of ping-pong handovers. However, delayed handovers may cause service drops. A smaller value of this parameter results in a higher probability of triggering inter-TRP handovers, a lower probability of service drops, and a higher probability of ping-pong handovers.
NRDUCellMultiTrp.PingPongHoRsrpThldOfs	Specifies the RSRP threshold offset for ping-pong handover prevention.	The value of this parameter plus that of the NRDUCellMultiTrp.TrpHoDeltaRsrp parameter is used as the threshold for a handover back to the original serving TRP within 2 seconds after a handover. The threshold for an inter-TRP handover is restored to the value of the NRDUCellMultiTrp.TrpHoDeltaRsrp parameter as long as this handover occurs more than 2 seconds after the previous handover.

3. The gNodeB updates the serving TRP.

The gNodeB uses the candidate TRP as the new serving TRP, which performs scheduling and allocates resources to the UE.

When TRPs experience interference and weak coverage, or overshoot-coverage TRPs exist in a cell, the following optimizations can be implemented.

- If TRPs experience interference or weak coverage, methods described in [Table 4-4](#) can be used to improve the inter-TRP handover reliability.

Table 4-4 Methods of improving the inter-TRP handover reliability when TRPs experience interference or weak coverage

Methods of Improving Inter-TRP Handover Reliability	Description
Increasing the number of times Media Access Control control elements (MAC CEs) are sent during an inter-TRP handover	This number can be increased by setting the NRDUCellMultiTrp.InterTrpHoMacCeTransTimes parameter to a larger value. In addition, the NRDUCellMultiTrp.TrpHoMacCeAdpTransNumOfs parameter can be used to configure the offset to the number of times MAC CEs are sent, so that MAC CEs can be sent for a larger number of times during an inter-TRP handover for a UE whose block error rate (BLER) is greater than the value of the NRDUCellMultiTrp.TrpHoMacCeAdpTransBlerThld parameter.
Increasing the offset to the power for sending MAC CEs during an inter-TRP handover	This offset can be increased by setting the NRDUCellMultiTrp.InterTrpHoMacCePwrOfs parameter to a larger value.
Filtering candidate TRPs based on SRS SINR	The NRDUCellMultiTrp.InterTrpHoSrsSnrThld and NRDUCellMultiTrp.InterTrpHoSrsSnrDiffThld parameters specify the SRS SINR threshold and SRS SINR difference threshold, respectively, which are regarded as SRS SINR-based evaluation conditions. During TRP selection for a handover, the gNodeB filters out a candidate TRP even if the TRP meets the SRS RSRP-based handover condition, when all of the following conditions are met: • NRDUCellMultiTrp.InterTrpHoSrsSnrThld > -20 • SRS SINR of the candidate TRP measured by the gNodeB < NRDUCellMultiTrp.InterTrpHoSrsSnrThld • SRS SINR difference between the serving and candidate TRPs ≥ NRDUCellMultiTrp.InterTrpHoSrsSnrDiffThld

- If overshoot-coverage TRPs exist in a cell (that is, the coverage areas of these TRPs exceed the planned ones), you are advised to optimize the coverage. If the coverage cannot be optimized in the short term, the parameters listed in [Table 4-5](#) can be used for performance optimization.

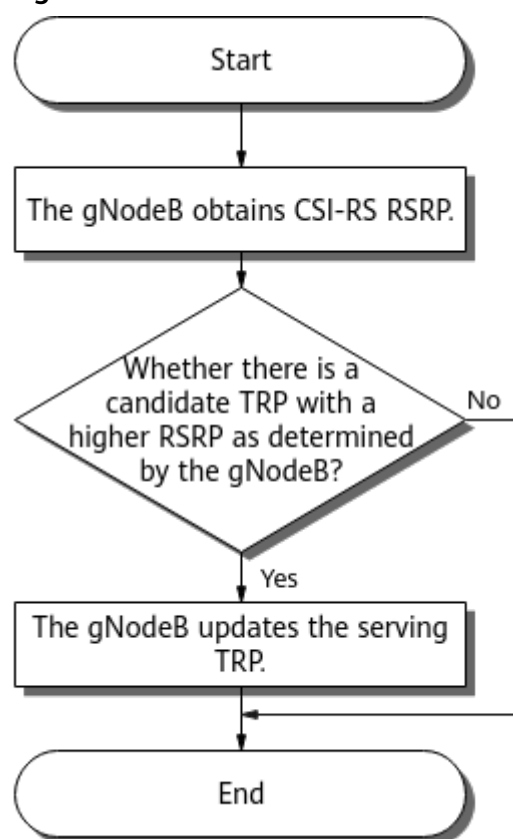
Table 4-5 Performance optimization parameters for overshoot-coverage TRPs

Object	Parameter ID	Meaning	Description
Target TRP	NRDUCellMultiTrp.TrpOvrCovTaThld	Specifies the overshoot-coverage threshold for a target TRP.	In TDD, if the overshoot-coverage distance of the target TRP relative to the serving TRP exceeds the value of the NRDUCellMultiTrp.TrpOvrCovTaThld parameter multiplied by 4.8 (m), the SRS RSRP threshold for a handover to this TRP changes to the sum of the NRDUCellMultiTrp.TrpHoDeltaRsrp and NRDUCellMultiTrp.TrpOvrCovHoRsrpThldOfs parameter values.
	NRDUCellMultiTrp.TrpOvrCovHoRsrpThldOfs	Specifies the offset to the overshoot-coverage threshold for a target TRP.	
Source TRP	NRDUCellMultiTrp.SrcTrpOvrCovTaThld	Specifies the overshoot-coverage threshold for a source TRP.	In TDD, if the overshoot-coverage distance of the source TRP relative to the target TRP exceeds the value of the NRDUCellMultiTrp.SrcTrpOvrCovTaThld parameter multiplied by 4.8 (m), the SRS RSRP threshold for a handover from the source TRP changes to the sum of the NRDUCellMultiTrp.TrpHoDeltaRsrp and NRDUCellMultiTrp.SrcTrpOvrCovHoRsrpThldOfs parameter values.
	NRDUCellMultiTrp.SrcTrpOvrCovHoRsrpThldOfs	Specifies the offset to the overshoot-coverage threshold for a source TRP.	
	TRP_OVER_COV_HO_OPT_SW option of the NRDUCellMultiTrp.MultiTrpAlgoSwitch parameter	TRP Overshoot-Coverage HO Opt Switch	TRP Overshoot-Coverage HO Opt Switch improves the accuracy of inter-TRP handovers. After this function is enabled, the source TRP's TA is checked for validity during an inter-TRP handover.

Inter-TRP Handover in CA SCells

When a UE for which asymmetric CA takes effect (that is, the UE uses different numbers of CCs in the uplink and downlink) moves between the TRPs for a hyper cell and it cannot send SRSs on SCCs, inter-TRP handover in CA SCells can be enabled by selecting the **SCell_INTER_TRP_HO_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter. This function relies on CSI-RS RSRP measurements. The specific process is shown in [Figure 4-4](#).

Figure 4-4 Process of an inter-TRP handover in a CA SCell



1. The gNodeB obtains CSI-RS RSRP.
The gNodeB sends CSI-RSs in each TRP and obtains CSI-RS RSRP measured and reported by the UE.
2. The gNodeB determines whether there is a candidate TRP with a higher RSRP.
The gNodeB determines whether there is a candidate TRP with a higher RSRP based on CSI-RS RSRP measured and reported by the UE.
If the number of consecutive times the RSRP of a candidate TRP meets a certain evaluation condition reaches the value of the **NRDUCellCsirs.InterTrpHoCsiRsrpEvalTimes** parameter, an inter-TRP handover is triggered. When a UE is handed over between TRPs for a hyper cell, the gNodeB does not need to send a handover command to the UE and the UE does not need to initiate an access procedure in the new serving TRP. The evaluation condition for triggering an inter-TRP handover is as follows: (CSI-RS RSRP of a candidate TRP – CSI-RS RSRP of the serving TRP) > (CSI-RS RSRP difference threshold + RSRP threshold offset for ping-pong handover prevention). Related parameters are described in [Table 4-6](#).

Table 4-6 Parameters related to inter-TRP handover in CA SCells

Parameter ID	Meaning	Description
NRDUCellCsirs.<i>InterTrpHoCsiRsrpEvalTimes</i>	Specifies the number of evaluation times for an inter-TRP handover based on CSI-RS RSRP measurements.	If the number of consecutive times a candidate TRP meets the evaluation condition reaches the value of this parameter, an inter-TRP handover is triggered. When signals fluctuate violently, you can set the parameter to a larger value to increase the number of evaluation times, avoiding ping-pong handovers between TRPs. When signals attenuate quickly, you can set the parameter to a smaller value to reduce the number of evaluation times, ensuring a prompt inter-TRP handover.
NRDUCellCsirs.<i>InterTrpHoDeltaCsiRsrpThld</i>	Specifies the CSI-RS RSRP difference threshold for an inter-TRP handover.	None
NRDUCellMultiTrp.<i>PingPongHoRsrpThldOfs</i>	Specifies the RSRP threshold offset for ping-pong handover prevention.	The value of this parameter plus that of the NRDUCellCsirs.<i>InterTrpHoDeltaCsiRsrpThld</i> parameter is used as the threshold for a handover back to the original serving TRP within 2 seconds after a handover. The threshold for an inter-TRP handover is restored to the value of the NRDUCellCsirs.<i>InterTrpHoDeltaCsiRsrpThld</i> parameter as long as this handover occurs more than 2 seconds after the previous handover.
NRDUCellCsirs.<i>CsiRsrpFilterCoef</i>	Specifies the CSI-RS RSRP filtering coefficient.	A smaller value of this parameter results in a longer interval between the time the inter-TRP CSI-RS meets the handover condition and the time a handover is triggered. It lowers the probability of ping-pong handovers but increases the service drop rate due to delayed handovers. A larger value of this parameter results in a shorter interval. It lifts the probability of a decrease in the service drop rate and the probability of ping-pong handovers.

Parameter ID	Meaning	Description
NRDUCellCsirs.CsirsRptBasedSccDeactivThld	Specifies the threshold for SCell deactivation based on CSI-RS measurement reports.	If the number of consecutive times the gNodeB fails to receive CSI-RS RSRP measurement results reported by UEs in an SCell is greater than or equal to this threshold, deactivation of this SCell is triggered. After the 5s deactivation timer expires, an SCell activation procedure is triggered again. For details about this procedure, see section "SCell Activation" in <i>Carrier Aggregation</i> . A smaller value of this parameter results in a higher probability of SCell deactivation. A larger value of this parameter results in a lower probability.

3. The gNodeB updates the serving TRP.
The gNodeB uses the candidate TRP as the new serving TRP, which performs scheduling and allocates resources to the UE.

4.1.4 Other Performance Optimization Functions

Table 4-7 describes other performance optimization functions provided by Hyper Cell.

Table 4-7 Other performance optimization functions

Function Name	Function Description	Function Switch
High-speed-specific interference randomization-based scheduling	With this function, the positions of resources to be scheduled for TRPs are staggered based on TRP ID mod 2 values, reducing mutual interference and improving uplink and downlink user-perceived rates.	NRDUCellAlgoSwitch.HighSpeedIntrfRandomSchSw
TRS configuration optimization	• If this function is disabled, TRS information is included in the RRC Reconfiguration message. • If this function is enabled, TRS information is included in the RRC Setup message, improving the access success rate of some UEs in high-speed scenarios.	TRS_CONFIG_OPT_SW option of the NRDUCellMultiTrp.MultiTrpAlgoSwitch parameter

Function Name	Function Description	Function Switch
PUSCH scrambling code optimization	This function allows UEs incompatible with UE-level scrambling codes to access the network.	NRDUCellMultiTrp.PuschScrambleOptSw
Inter-TRP RB resource sharing	This function is recommended in low-frequency TDD hyper cells. For details about inter-TRP RB resource sharing, see <i>Channel Management</i> .	INTER_TRP_RB_SHARE_POLICY_SW option of the NRDUCellPucch.PucchAlgoExtSwitch parameter
CA flexible networking	This function allows downlink intra-FR inter-band CA to take effect between hyper cells working on different frequencies or between hyper cells and common cells, increasing downlink user-perceived rates. This function takes effect only after the SCell_INTER_TRP_HO_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter is also selected. Selecting this option enables inter-TRP handover in CA SCells based on TRP-specific resource allocation for CSI-RS beam sweeping and CSI-RS RSRP measurements. For details about inter-TRP handover in CA SCells, see Inter-TRP Handover in CA SCells in 4.1.3 Inter-TRP Handover.	CA_FLEX_NETWORKING_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter
CA enhancement in hyper cell co-coverage scenarios or combined cell co-coverage scenarios	This function increases the average number of downlink CA UEs that use the local cell as their PCell and have activated SCells (indicated by N.User.CA.PCell.DL.SCellAct.Avg) and the average downlink throughput of CA UEs in scenarios (referred to as 3CC CA or massive CA co-coverage scenarios hereinafter) where all of the following conditions are met: - CA flexible networking is enabled. - Downlink intra-band CA or downlink intra-FR inter-band CA involving three or more carriers takes effect. - The PCell covers the same area as two or more SCells.	HYPER_CELL_CO_COV_CA_ENH_SW option of the gNodeBParam.NetworkingOptionOptSw parameter

Function Name	Function Description	Function Switch
CA flexible data split	This function enables the base station to flexibly determine when to split data, based on the packet arrival time and UE scheduling time, after downlink intra-FR inter-band CA takes effect between hyper cells working on different frequencies or between hyper cells and common cells. This improves the accuracy of data split, and thus increases the downlink user-perceived rates of CA UEs.	CA_FLEX_DATA_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter
Inter-TRP F4 HARQ resource sharing	In a multi-TRP cell, UEs that transmit HARQ feedback exceeding two bits can use more format-4 resources for HARQ feedback, which increases the average downlink throughput of these UEs.	INTER_TRP_F4_HARQ_SHARE_SW option of the NRDUCellPucch.PucchAlgoExtSwitch parameter
Adaptive CSI-RS beam sweeping for CA UEs	This function enables the gNodeB to adaptively stop CSI-RS beam sweeping on some carriers for specific CA UEs, reducing the number of aggregated carriers as required and preventing UE exceptions like UE reestablishment and service drops when the following conditions are met: (1) Inter-TRP handover in CA SCells is triggered based on CSI-RS RSRP measurements. (2) The number of beams used for CSI-RS for beam measurement (CSI-RS for BM) beam sweeping exceeds the UE capability indicated by <i>maxNumberSSB-CSI-RS-ResourceOneTx</i> and <i>maxNumberCSI-RS-Resource</i> IEs.	CA_CSIRS_BM_SWEEP_ADPT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter

4.2 Network Analysis

4.2.1 Benefits

- After the Hyper Cell function is enabled, the number of handovers decreases when UEs move between TRPs for a hyper cell. This prevents service interruptions or decreases in UE throughput caused by frequent handovers. The number of handovers reduced correlates with the number of combined TRPs. The more combined TRPs, the fewer the handovers. With common

channel joint transmission, the interference on common channels decreases compared with when TRPs are not combined.

- When the FSD transmission mode is used for hyper cells, the uplink and downlink scheduling and inter-TRP resource allocation of hyper cells become more flexible, and the performance of UEs in overlapping areas is further improved compared with those using the basic transmission mode. If inter-TRP interference, weak coverage, or overshoot coverage exists and coverage optimization cannot be performed in the short term, the FSD transmission mode is recommended.
- When CA flexible networking is enabled, CA combinations of more networking types are supported, increasing downlink throughput. In addition, peak downlink data rates of CA UEs in hyper cells increase.
- When adaptive CSI-RS beam sweeping for CA UEs is enabled, the base station employs a conservative policy on the number of carriers to prevent issues like RRC reestablishment and service drops, which can occur due to limited UE capability during inter-TRP handover in CA SCells.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

- [Table 4-8](#) describes the network impacts after a hyper cell is formed.

Table 4-8 Network impacts after a hyper cell is formed

Item	Impact
Number of UEs supported by a cell	As each TRP for a hyper cell consumes user instance resources, the number of UEs that a hyper cell supports decreases.
Cell traffic volume	• After a hyper cell is formed, the number of UEs that the cell needs to serve increases, which results in increased consumption of common channel resources (such as SRS resources), and decreased cell traffic volume in heavy-load scenarios. When common channel resources are insufficient, the service drop rate increases. • After a hyper cell is formed, handovers between common cells are triggered by SRS measurements of the base station instead of SSB measurement reports sent by UEs. Therefore, the handover points and UE distribution in common cells slightly change and the interference to UEs also changes. In addition, the uplink and downlink packet error rates (PERs) and service drop rate of the hyper cell fluctuate.
Service drop rate	
Uplink and downlink PERs	

Item	Impact
PRBs available for traffic channels	With common channel joint transmission, each TRP for a hyper cell has the same number of SSBs. If a TRP has a smaller number of SSBs than other TRPs before the combination, its SSB resource consumption increases and the PRBs available for traffic channels slightly decrease after the combination. However, SSB's interference to PDSCH is reduced.
Access success rate	After a hyper cell is formed, the preambles received by multiple TRPs are processed in a unified manner, and the slots for sending Msg2 in the hyper cell may be different from those for sending Msg 2 in a common cell. As the BLER varies in different slots, the access success rate is affected.
SRS SINR	After a hyper cell is formed, the SRS resources of all UEs in the cell are allocated in a unified manner. The SRS interference between TRPs may decrease. When this interference decreases, SRS SINR increases.

- After a hyper cell is formed and the FSD transmission mode is used, inter-TRP PUSCH, PDSCH, and PDCCH resources are allocated more flexibly than when the basic transmission mode is used. Since scheduling of UEs changes, the following impacts may occur:
 - The CPU usage may increase.
 - The interference to UEs in the overlapping area decreases, the average MCS index of uplink and downlink UEs in the cell increases, the bit error rate decreases, the PDCCH CCE aggregation level decreases, and the service drop rate decreases.
 - In the overlapping area, UEs are scheduled separately in the frequency domain. In scenarios with the same traffic volume, the PRB usage of uplink and downlink traffic channels and the PDCCH CCE usage of the cell may decrease.
 - PDCCH resources are scheduled in frequency division mode for UEs in the overlapping area between TRPs. As a result, the available CCE resources may decrease.
 - If this mode is enabled in low inter-TRP interference and heavy-load scenarios, the number of RBs that can be scheduled may decrease due to frequency-domain staggering in the overlapping area. As a result, the cell-level uplink and downlink throughputs and the UE-level uplink and downlink throughputs decrease.
 - If the **NRDUCellMultiTrp.DlIntrfTrpSrsRsrpDiffThld** parameter is set to an excessively large or small value, interfering TRPs may not be accurately identified, and the overall cell performance may deteriorate. If this parameter is set to an excessively small value, interfering TRPs cannot be identified, and effective interference avoidance scheduling cannot be performed to improve the performance of UEs in the overlapping area. If this parameter is set to an excessively large value, the available resources for scheduling may be limited due to excessive avoidance in medium- and heavy-load scenarios. As a result, the risks of

- decreasing cell- and UE-level uplink and downlink throughputs increase. The default value is recommended.
- If inter-BBP scenarios are involved and the number of SSB beams varies under different TRPs, the maximum number of SSB beams of each TRP in the cell is used. As a result, the amount of PDSCH resources that can be scheduled may decrease.
 - After a hyper cell is formed and CA flexible networking is enabled:
 - As for **N.QosFlow.AbnormRel.RNL.RlcReset** and **N.QosFlow.AbnormRel.RNL.ULSyncFail**:

If the cell is a high-speed one, the downlink scheduling capability is enhanced for CA UEs due to data distribution to SCCs. Therefore, after CA takes effect in a high-BLER scenario, the maximum number of downlink RLC retransmissions is less likely to reach than when only one carrier is used in the scenario. If resynchronization occurs in the uplink when the time alignment timer (specified by the **NRDUCellULTaConfig.ULTimeAlignmentTimer** parameter) expires, the value of **N.QosFlow.AbnormRel.RNL.RlcReset** will decrease due to RLC resets in the cell, the value of **N.QosFlow.AbnormRel.RNL.ULSyncFail** will increase due to uplink resynchronization failures, but the total number of abnormal releases will remain unchanged.
 - For details about other network impacts of CA, see the "Impacts" section for intra-FR inter-band CA in *Carrier Aggregation*.
 - After a hyper cell is set up, if inter-TRP F4 HARQ resource sharing is enabled, the cell can support more UEs with HARQ feedback of more than two bits. The proportion of PUCCH allocation failures decreases, downlink throughput increases, but the probability of inter-TRP F4 HARQ resource conflicts may increase, increasing the downlink bit error rate.
 - When adaptive CSI-RS beam sweeping for CA UEs is enabled, the number of aggregated carriers may decrease.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-030205	Hyper Cell	NR0S00HPCL00	per Cell

Feature ID	Feature Name	Model	Sales Unit
For the license control item NR0S00HPCL00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the NRDUCell.NrDuCellNetworkingMode parameter value cannot be changed to HYPER_CELL due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.			

The sales unit "per Cell" indicates the number of TRPs. Specifically, the number of NR0S00HPCL00 license units required by a hyper cell equals the number of TRPs for the hyper cell. For other features enabled for TRPs for a hyper cell, if their license sales unit is "per Cell", the number of required license units also equals the number of TRPs for the hyper cell.

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TRS Allocation Strategy	NRDUCellCsirs.TrsAllocationStrategy	None	It is recommended that this parameter be set to MULTI_TRS for a hyper cell. Setting this parameter to SINGLE_TRS is recommended only when there are UEs with compatibility issues on the network. Otherwise, the service drop rate may slightly increase.
Low-frequency TDD	TRS rate matching	TRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	Multiple sets of TRS resources are allocated to UEs in a hyper cell, and the symbols for transmitting TRSs cannot be used to transmit PDSCH data. Therefore, TRS rate matching must be enabled before enabling Hyper Cell.
Low-frequency TDD	Downlink interference randomization-based scheduling	DL_INTRF_RANDOM_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeEnhSwitch parameter	<i>Downlink Scheduling</i>	High-speed-specific interference randomization-based scheduling (controlled by NRDUCellAlgoSwitch.HighSpeedIntrfRandomSchSw) requires that at least one of the following function switches be turned on: DL_INTRF_RANDOM_SW of the NRDUCellAlgoSwitch.AdaptiveEdgeEnhSwitch parameter NRDUCellAlgoSwitch.UlIntrfRandomSchSwitch
Low-frequency TDD	Uplink interference randomization-based scheduling	NRDUCellAlgoSwitch.UlIntrfRandomSchSwitch	<i>Uplink Scheduling</i>	

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	High-speed-specific interference randomization-based scheduling can be enabled only in hyper cells with the NRDUCell.HighSpeedFlag parameter set to HIGH_SPEED .
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	
Low-frequency TDD	Full-bandwidth initial BWP configuration	INIT_BWP_FULL_BW_SW option of the NRDUCellAlgoSwitch.h.BwpConfigPolicySwitch parameter	<i>5G Networking and Signaling</i>	The TRS_CONFIG_OPT_SW option of the NRDUCellMultiTrp.MultiTrpAlgoSwitch parameter can be selected only when the INIT_BWP_FULL_BW_SW option of the NRDUCellAlgoSwitch.BwpConfigPolicySwitch parameter is selected.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	CA flexible networking can be enabled only in hyper cells.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Multi-TRS optimization in TRP scenarios can be enabled only in hyper cells when the TRS resource allocation policy is set to MULTI_TRS .
Low-frequency TDD	TRS Allocation Strategy	NRDUCellCsirs.TrsAllocationStrategy set to MULTI_TRS	<i>Hyper Cell</i>	

4.3.2.2 Mutually Exclusive Functions

Some features or functions may not be supported when certain NR DU cell networking modes are used. For details about the cell requirements of these features or functions, see section "Requirements" in the corresponding feature documents.

- Basic functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Format-1 HARQ-ACK resource scheduling in RB saving mode	HARQ_ACK_RB_SAVING_SCH_SW option of the NRDUCellPucch.SrResoureAlgoSwitch parameter	<i>Channel Management</i>	Format-1 HARQ-ACK resource scheduling in RB saving mode cannot be enabled in hyper cells.
Low-frequency TDD	Downlink additional DMRS position adaptation	DL_ADD_DMRS_POS_ADAPT_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	None	Downlink additional DMRS position adaptation cannot be enabled in hyper cells.
Low-frequency TDD	Admission control	QOS_FLOW_ADMISSION_SW and QOS_FLOW_PREEMPTION_SW options of the NRDUCellAlgoSwitch.RacAlgoSwitch parameter	<i>Admission Control</i>	Admission control cannot be enabled in hyper cells.
Low-frequency TDD	SRS interference coordination based on self-contained and uplink slots	SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Channel Management</i>	SRS interference coordination based on self-contained and uplink slots cannot be enabled in hyper cells.
Low-frequency TDD	Optimization of non-contention-based RA beyond cell radius	CONTENTION_FREE_RA_OPT_SW option of the NRDUCellPrach.RachAlgoOptSwitch parameter	<i>Random Access Control</i>	Optimization of non-contention-based RA beyond cell radius cannot be enabled in hyper cells.
Low-frequency TDD	CSI-RS for CM reuse 3	CSIRS_REUSE3_SW option of the NRDUCellCsirs.CsSwitch parameter	<i>Channel Management</i>	CSI-RS for CM reuse 3 cannot be enabled in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Narrowband TRS rate matching	NARROW_BAND_TRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	None	Narrowband TRS rate matching cannot be enabled in hyper cells with the BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter selected.
Low-frequency TDD	UE-specific CSI-RS reconfiguration reduction	USER_CSI_RECFG_TIMES_RDC_SW option of the NRDUCellCsirs.CsirsExtSwitch parameter	<i>Channel Management</i>	UE-specific CSI-RS reconfiguration reduction cannot be enabled in hyper cells.

- Functions related to energy saving

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	For macro base stations, Hyper Cell and RF channel intelligent shutdown can be both enabled in high-speed scenarios only when the HIGH_SPEED_CELL_MUTING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter is selected. These functions can also be both enabled in low-speed scenarios. After half of RF channels are shut down in low-speed scenarios, the working transmit and receive channel combinations must be compatible with Hyper Cell. For details about the transmit and receive channel combinations supported by Hyper Cell, see 4.3.3 Hardware . For LampSite base stations, Hyper Cell and RF channel intelligent shutdown can be both enabled.
Low-frequency TDD	Energy-efficiency-based inter-frequency handover	ENERGY EFFICIENCY_BASED_HO option of the NRCCellAlgoSwitch.InterFreqHoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Energy-efficiency-based inter-frequency handover cannot be enabled in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Energy-efficiency-based coordinated scheduling	ENERGY_EFF_BASED_COLLAB_SCH_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Energy-efficiency-based coordinated scheduling cannot be enabled in hyper cells.
Low-frequency TDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Uplink symbol power saving cannot be enabled in hyper cells.
Low-frequency TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Dynamic RF channel shutdown cannot be enabled in hyper cells.
Low-frequency TDD	Common-channel dynamic shutdown	NRDUCellPowerSaving.ComChnDynShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	Common-channel dynamic shutdown cannot be enabled in hyper cells with the NRDUCellPowerSaving.PowerSavingType parameter set to TTI_RF_SHUTDOWN or RF_CHN_DYN_SHUTDOWN .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inactive RA-SDT	UE_PWR_SAVING_ENHANCE_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter and the RRC_INACTIVE_RA_SDT_SW option of the NRCellAlgoSwitch./InactiveStrategySwitch parameter	<i>UE Power Saving</i>	The RRC_INACTIVE_RA_SDT_SW option of the NRCellAlgoSwitch./InactiveStrategySwitch parameter cannot be selected in hyper cells.
Low-frequency TDD	LTE and NR bundled scheduling	LNR_BUNDLED_SCHEDULING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	LTE and NR bundled scheduling cannot be enabled in hyper cells.
Low-frequency TDD	LTE and NR intelligent carrier shutdown	LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	LTE and NR intelligent carrier shutdown cannot be enabled in hyper cells.
Low-frequency TDD	Inter-TTI time-frequency-power collaboration	INTER_TTI_TFP_COOPERATION_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter.	None	Inter-TTI time-frequency-power collaboration cannot be enabled in hyper cells.
Low-frequency TDD	Backup power saving policy	NRDUCellPowerSaving.BackupPwrSavingPolicy	<i>Green Site</i>	The parameter specifying the backup power saving policy must be set to PRECISE_BACKUP_POLICY1 for hyper cells.
Low-frequency TDD	Intelligent power control energy saving	SMART_PWR_CTRL_ENERGY_SAVE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Intelligent power control energy saving cannot be enabled in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Common-channel dynamic shutdown cannot be enabled for hyper cells.

- Functions related to MIMO

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	Distributed massive MIMO cannot be enabled in hyper cells.
Low-frequency TDD	MetaAAU beam sensing	NRDUCellBeamAlgo.BeamPerceiveMode set to CENTRALIZED_MODE	<i>MIMO (TDD)</i>	MetaAAU beam sensing cannot be enabled in hyper cells.
Low-frequency TDD	Blind SRS interference suppression	SRS_BLIND_IS_SW option of the NRDUCellDlMimo.HighLayerMuMimoSw parameter	<i>MIMO (TDD)</i>	Blind SRS interference suppression cannot be enabled in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS interference coordination	SRS_IC_SW option of the NRDUCellSrs.SrsDetectionAlgoSwitch parameter, the SRS_JOINT_PC_SW option of the NRDUCellUIPConfig.UlPwrCtrlAlgoSwitch parameter, and the BEAM_SPECIFIC_SRS_DENOISE_SW option of the NRDUCellSrs.SrsDetectionAlgoSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	When the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL , the following three sub-functions of SRS interference coordination cannot be enabled: SRS IC, SRS joint power control, and SRS beam-specific noise reduction.
Low-frequency TDD	Interference rejection precoding for downlink MU-MIMO in NLOS scenarios	DL_MU_NLOS_IR_PRECODING_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	Interference rejection precoding for downlink MU-MIMO in NLOS scenarios cannot be enabled in hyper cells.
Low-frequency TDD	Adaptive CSI-RS multi-beam adjustment	NRDUCellCsirs.CsirsCellResourceNum set to FD_RESOURCE	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	In 32T32R+64T64R mixed configuration scenarios, adaptive CSI-RS multi-beam adjustment cannot be enabled.
Low-frequency TDD	PUSCH coordinated power control	PUSCH_COORD_PWR_CTRL_SW option of the NRDUCellUIPConfig.UlPwrCtrlAlgoExtSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	PUSCH coordinated power control cannot be enabled in hyper cells.

- Functions related to carrier management

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-FR CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Inter-FR CA cannot be enabled in hyper cells.
Low-frequency TDD	Downlink fast inter-gNodeB CA	INTER_GNB_ULTRA_LOW_LAT_SPLT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>CA Experience Improvement</i>	If the CA_FLEX_DATA_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter is deselected, downlink fast inter-gNodeB CA cannot be enabled in hyper cells.
Low-frequency TDD	Virtual grid-based multi-frequency coordination	NR_IF_HO_MEAS_QTY_PRED_FUN_SW option of the NRCCellSmartMultiCarr.NrMultiCarrierAlgoSwitch parameter	<i>Virtual Grid-based Multi-Frequency Coordination</i>	Virtual grid-based multi-frequency coordination cannot be enabled in hyper cells.
Low-frequency TDD	CCE-based data split stop for CA UEs	CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	CCE-based data split stop for CA UEs cannot be enabled in hyper cells.
Low-frequency TDD	Multi-frequency beam coordination	MULTI_FREQ_BEAM_COORD_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	<i>Multi-Frequency Smart Selection</i>	Multi-frequency beam coordination cannot be enabled in hyper cells.
Low-frequency TDD	CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_MD_OPT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	CA Performance Boost Based on Multi-dimensional Optimization cannot be enabled in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SCC coverage expansion	SA networking: SA_SCC_COV_THLD_ADAPT_SW option of the NRCellAlgoSwitch . MultiFreqAlgoSwitch parameter NSA networking: NSA_SCC_COV_THLD_ADAPT_SW option of the NRCellAlgoSwitch . MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	SCC coverage extension cannot be enabled in hyper cells.
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_SMART_SEL_SW option of the NRCellAlgoSwitch . MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	Experience-based smart carrier selection cannot be enabled in hyper cells.
Low-frequency TDD	Single DCI	DL_SINGLE_DCI_SW option of the NRDUCellCarrMgmt . CaEnhancedAlgoSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	Single DCI cannot be enabled in hyper cells.
Low-frequency TDD	Allow Virtual Grid Model Build Flag	VG_MODEL_ALLOW_BUILD_FLAG option of the NRCellFreqRelation . AggregationAttribute parameter	<i>Virtual Grid-based Multi-Frequency Coordination</i>	None
Low-frequency TDD	DL Intra-Freq Spct Eff Mdl Build Allowed	DL_INTRAF_SE_MODEL_BUILD_ALLOW option of the NRCellSmartMulti-Carr . ModelBuildingSwitch parameter	<i>Virtual Grid-based Multi-Frequency Coordination</i>	None

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Allow Virtual Grid Model Build Flag	VG_MODEL_ALLOW_BUILD_FLAG option of the NRCellEutranNFreqAggregationAttribute parameter	<i>Interoperability Between E-UTRA N and NG-RAN in Connected Mode</i>	None
Low-frequency TDD	Intra-frequency SRS beam prediction model	INTRA_SRS_MODEL_BUILD_ALLOW option of the NRCellSmartMultiCarr.ModelBuildingSwitch parameter	<i>Virtual Grid-based Multi-Frequency Coordination</i>	None

- Functions related to coordination

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	LTE TDD and NR Flash Dynamic Power Sharing cannot be enabled in hyper cells.
Low-frequency TDD	Co-MM	INTRA_GNB_SU_CO_MM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	Co-MM cannot be enabled in hyper cells.
Low-frequency TDD	Intra-base-station UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	(TDD) Intra-base-station UL CoMP cannot be enabled in hyper cells using the mixed configuration of 2T2R+4T4R.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	(TDD) Intra-base-station DL CoMP cannot be enabled in hyper cells using the mixed configuration of 2T2R+4T4R.
Low-frequency TDD	Inter-base-station UL CoMP	INTER_GNB_UL_CoMP_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	None	Inter-base-station UL CoMP cannot be enabled in hyper cells.
Low-frequency TDD	Layer increase for joint transmission in indoor cells	INDOOR_JT_LAYER_INC_SW option of the NRDUCellComp.DlCompAlgoSwitch parameter	None	Layer increase for joint transmission in indoor cells cannot be enabled in hyper cells.

- Functions related to interference reduction (remote interference)

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INTRF_AVOID_SW and SRS_INTRF_COORD_S_SLOT_SW options of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	Proactive avoidance of remote interference cannot be enabled in hyper cells.
Low-frequency TDD	Remote interference resistance for PUSCH demodulation	RIM_PUSCH_DEMOD_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	Remote interference resistance for PUSCH demodulation cannot be enabled in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS remote interference avoidance	SRS_RIM_INTRF_AVOID_SW option of the NRDUCellSrs.SrsAlgorithmExtSwitch parameter	None	SRS remote interference avoidance cannot be enabled in low-speed hyper cells (for which the NRDUCell.HighSpeedFlag parameter is set to LOW_SPEED).
Low-frequency TDD	Interference Avoid Detect Period Num	NRDUCellRimConfig.IntrfAvoidDetectPrdNum	<i>Remote Interference Management (Low-Frequency TDD)</i>	If remote interference adaptive avoidance or enhanced remote interference adaptive avoidance needs to be enabled in hyper cells, the NRDUCellRimConfig.IntrfAvoidDetectPrdNum parameter cannot be set to 0 .
Low-frequency TDD	RIM-RS Type1 Intrf Thld	NRDUCellRimConfig.RimRsType1IntrfThld	<i>Remote Interference Management (Low-Frequency TDD)</i>	If remote interference adaptive avoidance or enhanced remote interference adaptive avoidance needs to be enabled in hyper cells, the NRDUCellRimConfig.RimRsType1IntrfThld parameter must be set to -50 .
Low-frequency TDD	Forced remote interference avoidance	FORCED_RIM_AVOID_SW option of the NRDUCellRimConfig.RimPolicySwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	If remote interference adaptive avoidance or enhanced remote interference adaptive avoidance needs to be enabled in hyper cells, the FORCED_RIM_AVOID_SW option of the NRDUCellRimConfig.RimPolicySwitch parameter cannot be selected.

- Functions related to interference reduction (other interference)

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Interference-based subband robust weights	INTRF_BASED_ROBUST_WEIGHT_SW option of the NRDUCellIntrfIdent.ChnMeasCtrlSw parameter	<i>Interference Avoidance</i>	Interference-based subband robust weights cannot be enabled in hyper cells.
Low-frequency TDD	Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW and INTER_GNB_DL_CS_CBF_SW options of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	Coordinated interference management cannot be enabled in hyper cells.
Low-frequency TDD	PUCCH interference avoidance	PUCCH_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter	None	PUCCH interference avoidance cannot be enabled in hyper cells.
Low-frequency TDD	Enhanced intra-frequency SSB interference avoidance	INTRA_FREQ_SSB_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	The inter-TRP SSB staggering function (controlled by the INTER_TRP_SSB_STAGGER_SWITCH option of the NRDUCellMultiTrp.MultiTrpAlgoSwitch parameter) of Hyper Cell is mutually exclusive with enhanced intra-frequency SSB interference avoidance.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced inter-frequency SSB interference avoidance	INTER_FREQ_SSB_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrFvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	The inter-TRP SSB staggering function (controlled by the INTER_TRP_SSB_STAGGER_SWITCH option of the NRDUCellMultiTrp.MultiTrpAlgoSwitch parameter) of Hyper Cell is mutually exclusive with enhanced inter-frequency SSB interference avoidance.
Low-frequency TDD	Interference avoidance for indoor cells	INDOOR_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.IndoorOutdoorCooperateSw parameter	<i>3D Networking Experience Improvement</i>	Interference avoidance for indoor cells cannot be enabled in hyper cells.
Low-frequency TDD	Enhanced interference avoidance for indoor cells	INDOOR_INTRF_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.IndoorOutdoorCooperateSw parameter	<i>3D Networking Experience Improvement</i>	Enhanced interference avoidance for indoor cells cannot be enabled in hyper cells.

- Functions related to RAN services (FWA)

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	FWA private line service assurance	PRIVATE_LINE_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	FWA private line service assurance cannot be enabled in hyper cells.
Low-frequency TDD	Downlink experience-based scheduling	DL_EXP_BASED_SCH_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	Downlink experience-based scheduling cannot be enabled in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	User PRB control for FWA	PRB_CTRL_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	User PRB control for FWA cannot be enabled in hyper cells.
Low-frequency TDD	SRS antenna selection for RedCap users in FWA scenarios	REDCAP_SRS_ANT_SWITCH_SW option of the NRDUCellRedCap.RedCapAlgoSwitch parameter	<i>FWA</i>	SRS antenna selection for RedCap users in FWA scenarios cannot be enabled in hyper cells.
Low-frequency TDD	Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDLRank.DlRankSelAlgoSwitch parameter	<i>FWA</i>	Downlink 8-layer transmission for a single FWA user cannot be enabled in hyper cells.

- Functions related to RAN services (others)

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink semi-persistent scheduling	DL_SPS_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter	<i>URLLC</i>	Downlink semi-persistent scheduling cannot be enabled in hyper cells.
Low-frequency TDD	SRS field strength-based positioning	SRS_RSRP_LOCATION_SW option of the NRCCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	SRS field strength-based positioning cannot be enabled in hyper cells.
Low-frequency TDD	U-TDOA-based positioning/TOA fingerprint-based positioning	UTDOA_SW option of the NRCCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	U-TDOA-based positioning/TOA fingerprint-based positioning cannot be enabled in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Positioning-dedicated SRS transmission in uplink slots	SRS_FOR_POSITION_ON_USLOT_SW option of the NRDUCellPosAlgo.LcsMeasureAlgoSw parameter	<i>Positioning</i>	Positioning-dedicated SRS transmission in uplink slots cannot be enabled in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multiple BWP1s for RedCap	NRDUCellRedCap.RedCapUeMaxAvail-Bandwidth set to a value greater than 20M	<i>RedCap</i>	For a hyper cell for which the REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter is enabled, the NRDUCellRedCap.RedCapUeMaxAvailBandwidth parameter must be set to 20M, 40M, or 60M if one of the following conditions is met: - The REDCAP_PUCCH_RES_GUAR_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter is deselected. - The REDCAP_PUCCH_RES_GUAR_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter is selected and at least one of the USER_NUM_BASED_BWP_ACTV_SW, PDCCH_CCE_BASED_BWP_ACTV_SW, and RESOURCE_BASED_BWP_ACTV_SW options of the NRDUCellRedCap.RedCapAlgoSwitch parameter is selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Scheduling in common PDCCH symbols for RedCap UEs	REDCAP_COMM_PDCCH_SYM_SCH_SW option of the NRDUCellRedCap.RedCapAlgoSwitch parameter	<i>RedCap</i>	Scheduling in common PDCCH symbols for RedCap UEs cannot be enabled in hyper cells.
Low-frequency TDD	Assistance TRS	ASSISTANCE_TRS_SW option of the NRDUCellUePwrSaving.IdleUePwrSavingSwitch parameter	<i>RedCap</i>	Assistance TRS cannot be enabled in hyper cells.

- Functions related to coverage improvement

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UL and DL Decoupling	NRDUCellAlgoSwitch.UIDecouplingSwitch	<i>UL and DL Decoupling</i>	UL and DL Decoupling cannot be enabled in hyper cells.
Low-frequency TDD	Super uplink with independent SUL deployment	SUPER_UL_TDM_SCH_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	Super uplink with independent SUL deployment cannot be enabled in hyper cells.
Low-frequency TDD	Super uplink with SUL and NR FDD co-deployment	FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	Super uplink with SUL and NR FDD co-deployment cannot be enabled in low-speed hyper cells (for which the NRDUCell.HighSpeedFlag parameter is set to LOW_SPEED).
Low-frequency TDD	Extended Cell Range	NRDUCell.CellRadius set to a value greater than 14500	<i>Extended Cell Range</i>	Extended Cell Range cannot be enabled in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Extended Cell Range of 90 km (TDD)	NRDUCell.CellRadius set to a value greater than 60000 and less than or equal to 90000	<i>Extended Cell Range</i>	Extended Cell Range of 90 km (TDD) cannot be enabled in hyper cells.

- Other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	5QI-specific UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCCellAlgoSwitch.MlbAlgoSwitch parameter selected, and NRCCellMlb.MlbTriggerMode set to SPECIFIED_5QI_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	5QI-specific UE-number-based connected mode MLB cannot be enabled in hyper cells.
Low-frequency TDD	DACQ	DACQ_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	None	Dynamic AMBR control by QCI (DACQ) cannot be enabled in hyper cells.
Low-frequency TDD	CSI-RS scheduling enhancement	CSIRS_SCH_ENH_SW option of the NRDUCellComp.DlCompAlgoSwitch parameter	None	CSI-RS scheduling enhancement cannot be enabled in hyper cells.
Low-frequency TDD	Dynamic SSB	DYNAMIC_SSB_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	None	Dynamic SSB cannot be enabled in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	FR1 Max Cell CSI-RS Port Num	NRDUCellCsirs.FR1MaxCellCsirsPortNum set to 32PORT	None	In low-frequency TDD, the NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter can be set to 32PORT only if the NRDUCell.HighSpeedFlag parameter is set to HIGH_SPEED and the NRDUCellTrp.TxRxMode parameter is set to 32T32R .
Low-frequency TDD	Flexible spectrum access	FLEX_SPCT_ACCESS_SW option of the NRDUCellCarrMgmt.MultiBandFlexSwitch parameter	<i>Uplink Flexible Spectrum Scheduling</i>	Flexible spectrum access cannot be enabled in hyper cells.
Low-frequency TDD	Percentage-based dynamic RB sharing among operators	RB_DYNAMIC_SHARING_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	Percentage-based dynamic RB sharing among operators cannot be enabled in hyper cells with the BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter selected.
Low-frequency TDD	Operator-specific limitation on the maximum number of available RBs	MAX_RB_LIMIT_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	Operator-specific limitation on the maximum number of available RBs cannot be enabled in hyper cells with the BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter selected.
Low-frequency TDD	SRS-based weight enhancement	JT_SRS_WEIGHT_ENH_SW option of the NRDUCellComp.DICompAlgoSwitch parameter	None	SRS-based weight enhancement cannot be enabled in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Distributed antenna weight optimization	DISTRIBUTED_ANT_WEIGHT_OPT_SW option of the NRDUCellPdsch.DlPdschAlgoSwitch parameter	None	Distributed antenna weight optimization cannot be enabled in hyper cells.
Low-frequency TDD	PMI-based weight optimization for distributed antennas	DISTRIBUTED_ANT_PMI_WT_OPT_SW option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter	None	PMI-based weight optimization for distributed antennas cannot be enabled in a hyper cell meeting the following conditions: (1) The NRDUCellTrpBeam.TrpAntType parameter setting is inconsistent among all TRPs for the hyper cell which are referenced by the NRDUCellTrp.NrDualTrpld parameter. (2) The CELL_LEVEL_SCRAMBLED_SW option of the NRDUCellMultiTrp.MultiTrpCsiRsAlgoSwitch for the hyper cell is selected.
Low-frequency TDD	JT CSI-RS Transmit Mode	NRDUCellComp.JtCsirsTxMode	None	The NRDUCellComp.JtCsirsTxMode parameter cannot be set to PERIODIC_TD_MODE_ABN or PERIODIC_TD_MODE_ALL in hyper cells using the mixed configuration of 4T4R and 8T8R.
Low-frequency TDD	Uplink RF combining noise suppression	RF_COMBINING_NOISE_SUPPR_SW option of the NRDUCellUISch.RfCombSwitch parameter	None	Uplink RF combining noise suppression cannot be enabled in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink demodulation enhancement for UEs	DL_UE_DEMODULATE_ENH_SW option of the NRDUCellDischAlgoSwitch parameter	None	Downlink demodulation enhancement for UEs cannot be enabled in hyper cells.
Low-frequency TDD	DL Blanking	NRDUCell.PeerCellSlotAssignment	<i>Standards Compliance</i>	When DL Blanking is enabled (by setting the NRDUCell.PeerCellSlotAssignment parameter to a value other than NOT_CONFIG), inter-TRP SSB staggering cannot be enabled.
Low-frequency TDD	Downlink experience satisfaction guarantee	DL_EXP_SATIS_GUAR_SW option of the NRDUCellFwaQos.SchAlgoSwitch parameter	<i>FWA</i>	Downlink experience satisfaction guarantee cannot work with Hyper Cell.
Low-frequency TDD	Uplink 4-layer transmission for a single FWA user	UL_SU_FOUR_LAYER_SW option of the NRDUCellUIRank.UlRankAlgoSw parameter	<i>FWA</i>	Uplink 4-layer transmission for a single FWA user cannot be enabled in hyper cells.
Low-frequency TDD	FWA device-pipe-based uplink dense codebooks	UL_DENSER_CODEBOOK_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	<i>FWA</i>	FWA device-pipe-based uplink dense codebooks cannot be enabled in hyper cells.
Low-frequency TDD	FWA device-pipe-based uplink high power	UL_HIGH_POWER_UE_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	<i>FWA</i>	FWA device-pipe-based uplink high power cannot be enabled in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR DU cell resource management between network slices	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.NetworkSliceAlgoSwitch parameter	<i>Network Slicing</i>	NR DU cell resource management between network slices cannot be enabled if the FSD transmission mode is used.
Low-frequency TDD	RFSP-group-based RB resource control	RB_DYNAMIC_CONTROL_SW option of the NRDUCellCommonSw.RfspAlgoSwitch parameter	<i>Resource Control in NSA Networking</i>	RFSP-group-based RB resource control cannot be enabled if the FSD transmission mode is used.
Low-frequency TDD	Dynamic RB control	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter	<i>URLLC</i>	Dynamic RB control cannot be enabled if the FSD transmission mode is used.
Low-frequency TDD	QCI-based resource control	RB_DYNAMIC_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>	QCI-based resource control cannot be enabled if the FSD transmission mode is used.
Low-frequency TDD	Operator-specific guarantee on the minimum number of available RBs	MIN_RB_GUARANTEE_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	Operator-specific guarantee on the minimum number of available RBs cannot be enabled if the FSD transmission mode is used.

4.3.2.3 Function Impacts

- Basic functions (uplink)

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Basic functions of uplink scheduling	None	<i>Uplink Scheduling</i>	Uplink common control channels and signals such as the PUCCH, PRACH, and SRS for the TRPs of a hyper cell are scheduled in a unified manner. The gNodeB allocates the same resources to different TRPs for such channels and signals. Uplink UE-dedicated signaling and data for the TRPs of a hyper cell are independently scheduled and allocated by the serving TRP.
Low-frequency TDD	Uplink interference randomization-based scheduling	NRDUCellAlgoSwitch.UlIntrfRandomSchSwitch	<i>Uplink Scheduling</i>	Uplink interference randomization-based scheduling between TRPs is only supported in a hyper cell.
Low-frequency TDD	Uplink DCI extension-based scheduling	DCI_EXTENSION_UL_CONFIG_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>Uplink Scheduling</i>	For a hyper cell with the NRDUCell.HighSpeedFlag parameter set to HIGH_SPEED, enabling uplink DCI extension-based scheduling in light-load scenarios may decrease the downlink cell throughput and downlink user-perceived rate. Therefore, it is recommended that this function be enabled in scenarios where uplink CCEs are severely limited, for example, in tunnels or periods with a large number of high-speed UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUCCH RB adaptation	PUCCH_RB RES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>	After PUCCH RB adaptation is enabled, the number of RBs reserved for the PUCCH in a hyper cell is greater than that in a common cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-characteristic-based SRS period adaptation	USER_CHARACTERISTICS_ADAPT_SWITCH option of the NRDUCells parameter	<i>Channel Management</i>	When a cell (such as a common cell, a hyper cell, or a combined cell) is set up on a base station, a combination of the adaptation scheme and static SRS period configuration is supported for cells served by the base station, but the adaptation scheme must be the same (either SRS period adaptation or UE-characteristic-based SRS period adaptation). Specifically, - When there are no other activated cells on the base station, the new cell can use static SRS period configuration, SRS period adaptation, or UE-characteristic-based SRS period adaptation. - When there are activated cells using SRS period adaptation on the base station, UE-characteristic-based SRS period adaptation cannot be enabled for the new cell. Otherwise, the cell setup fails. To enable UE-characteristic-based SRS period adaptation for the new cell, perform the following operations: First, deactivate all activated cells using SRS period adaptation on the base station. Then, enable UE-characteristic-based SRS period adaptation or static SRS period configuration for the

RAT	Function Name	Function Switch	Reference	Description
				deactivated cells, and activate them. Lastly, enable UE-characteristic-based SRS period adaptation for the new cell. When there are activated cells using UE-characteristic-based SRS period adaptation on the base station, SRS period adaptation cannot be enabled for the new cell. Otherwise, the cell setup fails. To enable SRS period adaptation for the new cell, perform the following operations: First, deactivate all activated cells using UE-characteristic-based SRS period adaptation on the base station. Then, enable SRS period adaptation or static SRS period configuration for the deactivated cells, and activate them. Lastly, enable SRS period adaptation for the new cell.
Low-frequency TDD	RRC-reconfiguration-based dynamic waveform switching	UL_WAVEFORM_ADAPT_SW option of the NRDUCellAdaptiveEdgeEnhSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	RRC-reconfiguration-based dynamic waveform switching does not take effect in low-speed hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TRS-based path loss measurement	TRS_PATH_LOSS_MEAS_SW option of the NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch parameter	None	TRS-based path loss measurement is not supported in hyper cells.
Low-frequency TDD	PRACH fixed delay compensation	NRDUCellTrpAlgo.FixedDelayCompensation	<i>Random Access Control</i>	If the difference between the fixed delay compensation configurations of the serving and neighboring TRPs of a hyper cell is large, UE performance will be affected. This occurs when the difference between the delay from the UE to the serving TRP and that from the UE to a neighboring TRP is large (for example, in an overshoot coverage scenario).
Low-frequency TDD	PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Uplink MU-MIMO is not supported in hyper cells in the FSD mode.

- Basic functions (downlink)

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CSI-RS resource management	None	<i>Channel Management</i>	TRS is staggered for the TRPs of a hyper cell based on TRP ID mod 6 values to prevent mutual interference. CSI-RS for CM is configured in the same manner as that before TRP combination.
Low-frequency TDD	Basic functions of downlink scheduling	None	<i>Downlink Scheduling</i>	Downlink common control channels and signals, such as the PBCH and SS, for the TRPs of a hyper cell are scheduled in a unified manner and occupy the same resources. The same information is transmitted for TRPs. Downlink UE-dedicated signaling and data for the TRPs of a hyper cell are independently scheduled and allocated by the serving TRP.
Low-frequency TDD	Downlink interference randomization-based scheduling	DL_INTRF_RANDOM_SW option of the NRDUCellAlgoSwitch . AdaptiveEdgeExpEnhSwitch parameter	<i>Downlink Scheduling</i>	Downlink interference randomization-based scheduling between TRPs is only supported in a hyper cell.
Low-frequency TDD	3D coverage pattern	NRDUCellTrpBeam.CoverageScenario	<i>Beam Management</i>	It is recommended that the SCENARIO_BEAM_DENSIFY_ALGO_SW option of the NRDUCellTrpBeam.ScenarioBeamAlgoSw parameter have identical configurations for all the TRPs of a hyper cell to prevent UE access exceptions and improve system reliability.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell-level scrambling	CELL_LEVEL_SCRAMBLED_SW option of the NRDUCellMultiTrp.MultiTrpCsirsAlgoSwitch parameter	None	For TDD: - In 2T2R+4T4R, 1T1R+2T2R, or 1T1R+4T4R mixed configuration scenarios, the cell-level scrambling switch must be off. - In 32T32R+64T64R mixed configuration scenarios, if the NRDUCellCsirs.CsirsCellResourceNum parameter is not set to 1_RESOURCE for the cell, the cell-level scrambling switch must be off.
Low-frequency TDD	Narrowband and TRS rate matching	NARROW_BAND_TRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	None	- The TRS_RATEMATCH_SW and NARROW_BAND_TRS_RATEMATCH_SW options of the NRDUCellPdsch.RateMatchSwitch parameter cannot be both selected for a hyper cell. - The NARROW_BAND_TRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter can be selected only when the NRDUCellCsirs.TrsPeriod parameter is set to MS10 or MS20 for a hyper cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink correlation acceleration	DL_CORR_ACCELERATION_SW option of the NRDUCellDL MIMO.DLMuMimoSch SupplementSw parameter	None	Downlink correlation acceleration is not supported in hyper cells in the FSD mode.
Low-frequency TDD	Downlink anti-interference scheduling	DL_ANTI_INTERFERENCE_SW option of the NRDUCellPdsch.DIPdschAlgoSwitch parameter	None	Downlink anti-interference scheduling is not supported in hyper cells in the FSD mode.
Low-frequency TDD	PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	PDSCH spatial multiplexing is not supported in hyper cells in the FSD transmission mode.
Low-frequency TDD	Smart frequency selective scheduling	SMART_FREQUENCY_SELECTIVE_SW option of the NRDUCellPdsch.DIPdschAlgoSwitch parameter	None	If the FSD transmission mode is used for a hyper cell and this option is selected, the PDSCH throughput may significantly decrease due to staggering of overlapping areas. Therefore, enabling smart frequency selective scheduling is not recommended.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Spectral-efficiency-based adaptive switching between SU-MIMO and MU-MIMO	DL_SE_SU_MU_MIMO_ADAPT_SW option of the NRDUCellDL MIMO.DLMuMimoSwitchSw parameter	<i>MIMO (TDD)</i>	Spectral-efficiency-based adaptive switching between SU-MIMO and MU-MIMO is not supported in hyper cells in the FSD transmission mode.

- Functions related to coordination

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station UL CoMP	INTRA_GNB_UL_COMPSW option of the NRDUCellA lgoSwitch.CompSwitch parameter	<i>CoMP</i>	Intra-base-station UL CoMP is supported only between TRPs for a hyper cell, but not between hyper cells.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JTSW option of the NRDUCellA lgoSwitch.CompSwitch parameter	<i>CoMP</i>	Intra-base-station DL CoMP is supported only between TRPs for a hyper cell, but not between hyper cells.
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellA lgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	In Hyper Cell networking, the gains of NR inter-carrier dynamic power sharing may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_POWER_ADAPT_SWITCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	The gains provided by NR multi-band power adaptive scheduling may decrease in Hyper Cell networking.

- Functions related to carrier management

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink intra-band CA	INTRA_BAND_UL_CA_SWITCH option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Uplink intra-band CA is not supported in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SWITCH option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	- If CA flexible networking (controlled by the CA_FLEX_NETWORKING_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter) is disabled, downlink intra-FR inter-band CA is not supported in hyper cells. - If CA flexible networking is enabled, downlink intra-FR inter-band CA is supported in hyper cells. However, the cells involved in downlink intra-FR inter-band CA cannot include both hyper cells and combined cells. (Combined cells are those with NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to BASIC_MODE or DPS_MODE.)
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SWITCH option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	The cells involved in downlink intra-band CA cannot include both hyper cells and combined cells. (Combined cells are those with NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to BASIC_MODE or DPS_MODE .)

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	3CC aggregation	NRDUCellCarrMgmt.CaDLMaxCNum set to DL3CC	<i>CA Experience Improvement</i>	The cells involved in 3CC aggregation cannot include both hyper cells and combined cells. (Combined cells are those with NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTrasMode set to BASIC_MODE or DPS_MODE .)

- Functions related to interference reduction

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Remote interference adaptive avoidance	RIM_ADAPT_AVOID_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	After remote interference adaptive avoidance is enabled, only RIM-RS transmission and detection take effect in hyper cells (so that interference avoidance can be triggered in common cells), while GP adjustment does not. Because hyper cells have high traffic load, their resource consumption can be reduced when GP adjustment does not take effect. (RIM-RS is short for remote interference management reference signal, and GP is short for guard period.)

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced remote interference adaptive avoidance	RIM_ADAPT_AVOID_ENH_SW option of the NRDUCellAlgoSwitch . <i>RimAlgoSwitch</i> parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	After enhanced remote interference adaptive avoidance is enabled, only RIM-RS transmission and detection take effect in hyper cells (so that interference avoidance can be triggered in common cells), while adjustment in the time domain range for scheduling UEs does not. Because hyper cells have high traffic load, their resource consumption can be reduced when adjustment in the time domain range for scheduling UEs does not take effect.
Low-frequency TDD	Remote interference resistance for random access	RA_REMOTE_INTRF_RESIST_SW option of the NRDUCellAlgoSwitch . <i>RimAlgoSwitch</i> parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	Remote interference resistance for random access takes effect at the cell level. In a cell with multiple TRPs, if interference that is present at a TRP meets the conditions for triggering this function, this function will be triggered for that TRP and for other TRPs, even though the other TRPs do not experience interference. Consequently, the number of UEs served by the other TRPs may decrease, accompanied by fluctuations in the values of access-related counters.

- Other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Clock out-of-synchronization detection	CLK_DETECT_SW option of the gNodeBParam.ClkOutofsyncDetectSwitch parameter	<i>Synchronization</i>	Clock out-of-synchronization detection does not take effect for hyper cells.
Low-frequency TDD	Air-interface-based network-wide synchronization deviation detection	GAP_BASED_SYNC_DIFF_DETECT_SW option of the gNodeBParam.SyncDiffDetectSwitch parameter	<i>Synchronization</i>	Air-interface-based network-wide synchronization deviation detection does not take effect for hyper cells.
Low-frequency TDD	Uplink fallback to LTE	UL_FALLBACK_TO_LTE_SWITCH option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Basics</i>	In NSA networking, if uplink fallback to LTE is enabled in a hyper cell, the number of fallbacks to the LTE network may decrease, and the number of UEs in weak coverage areas served by the NR cell may increase. An increase in the number of UEs in weak coverage areas will cause the uplink and downlink cell throughput to decrease and the service drop rate to increase. It is recommended that the threshold for uplink fallback to LTE be increased for optimization.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Baseband shutdown for energy saving	BB_SHUTDOWN_POWERSAVING_SW option of the gNodeBParam.BaseStationDeepDormancySw parameter	<i>Energy Conservation and Emission Reduction</i>	If carrier shutdown takes effect for the TRPs of a hyper cell, the baseband processing resource blocks serving the TRPs cannot be shut down. If deep dormancy takes effect for the TRPs, the baseband processing resource blocks can be shut down.
Low-frequency TDD	Adaptive baseband shutdown	BBP.CELLEDPLOY set to ADAPT_BB_P_SHUTDOWN	<i>Energy Conservation and Emission Reduction</i>	If carrier shutdown takes effect for the TRPs of a hyper cell, the baseband processing resource blocks serving the TRPs cannot be shut down. If deep dormancy takes effect for the TRPs, the baseband processing resource blocks can be shut down.
Low-frequency TDD	Power saving first	BBP.CELLEDPLOY set to POWERSAVING	<i>Energy Conservation and Emission Reduction</i>	If carrier shutdown takes effect for the TRPs of a hyper cell, the baseband processing resource blocks serving the TRPs cannot be shut down. If deep dormancy takes effect for the TRPs, the baseband processing resource blocks can be shut down.
Low-frequency TDD	Simulated load	STRNRDUCELLSIMULATEDLOAD	None	The simulated load function is not supported in hyper cells.
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapBasebandFunSw parameter	<i>RedCap</i>	When RedCap Introduction is enabled, the NRDUCellBwp.DlMinCarrierBw parameter can only be set to 20M for a hyper cell with the downlink bandwidth greater than 20 MHz.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Super uplink with SUL and NR FDD co-deployment	FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	Super uplink with SUL and NR FDD co-deployment is not supported in high-speed hyper cells (for which the NRDUCell.HighSpeedFlag parameter is set to HIGH_SPEED).
Low-frequency TDD	Format-3 and Format-4 ACK RB resource adaptation	F3_AND_F4_ACK_RB_RESOURCES_ADAPT_SW option of the NRDUCellPucch.PucchAlgoExtSwitch parameter	<i>Channel Management</i>	When inter-TRP format-4 resource sharing for HARQ feedback is enabled together with format-3 and format-4 ACK RB resource adaptation, only format-3 and format-4 ACK RB resource adaptation takes effect.
Low-frequency TDD	Adaptive switching between hyper cell mode and combined cell mode	HYPERCELL_AND_COMBINECELL_ADPT option of the NRDUCellMultiTrp.MultiTrpAlgoSwitch parameter	None	Adaptive switching between hyper cell mode and combined cell mode is not supported in hyper cells in the FSD transmission mode.
Low-frequency TDD	High Speed Intrf Random Sch PRB Thld	NRDUCellHighSpeedMob.HSIntrfRandomSchPrbThld	<i>High Speed Mobility</i>	It is not recommended that NRDUCellHighSpeed-Mob.HSIntrfRandomSchPrbThld be set to 100 for hyper cells in FSD mode.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RB Estimate Intrf Random PRB Thld	- NRDUCellHighSpeedMob.<i>ULRbEstIntrfRandomPrbThld</i> - NRDUCellHighSpeedMob.<i>DLRbEstIntrfRandomPrbThld</i>	<i>High Speed Mobility</i>	It is not recommended that NRDUCellHighSpeedMob.<i>ULRbEstIntrfRandomPrbThld</i> and NRDUCellHighSpeedMob.<i>DLRbEstIntrfRandomPrbThld</i> be set to 100 for hyper cells in FSD mode.
Low-frequency TDD	Uplink basic preallocation	UL_BASIC_PREALLOCATION_SWITCH option of the NRDUCellPusch.<i>ULPreallocationSwitch</i> parameter	<i>Uplink Scheduling</i>	If the FSD transmission mode is used for a hyper cell and uplink basic preallocation is enabled, the number of preallocation times and the average uplink MCS index may decrease due to staggering of overlapping areas.
Low-frequency TDD	Uplink smart preallocation	UL_SMART_PREALLOCATION_SWITCH option of the NRDUCellPusch.<i>ULPreallocationSwitch</i> parameter	<i>Uplink Scheduling</i>	If the FSD transmission mode is used for a hyper cell and uplink smart preallocation is enabled, the number of preallocation times and the average uplink MCS index may decrease due to staggering of overlapping areas.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

 NOTE

The inter-TRP SSB staggering function controlled by the **INTER_TRP_SSB_STAGGER_SWITCH** option of the **NRDUCellMultiTrp.MultiTrpAlgoSwitch** parameter is supported only by macro base stations. Other functions of this feature are supported by both macro and LampSite base stations.

Boards

All NR-capable main control boards can be used.

In TDD, the baseband processing unit (BBP) must be a UBBPg series or later board or UBBPfw1. The configuration requirements are described in [Table 4-9](#).

Table 4-9 (TDD) Configuration requirements for BBPs

BBP	Dimension	Configuration Requirement
UBBPfw1	N/A	- Each BBP supports TRPs for only one hyper cell and all TRPs for the hyper cell must be configured on one BBP. - TRPs for hyper and common cells cannot be configured on the same BBP.
UBBPg series or later	Number of cells	- In scenarios where inter-board deployment exists, a BBP supports a maximum of three combined cells or hyper cells. If three hyper cells have been configured with the use of inter-board deployment, a fourth hyper cell cannot be configured. - In scenarios where inter-board deployment does not exist, a BBP supports a maximum of six combined cells or hyper cells.
	Number of TRPs	If a BBP serves TRPs for hyper cells, the total number of TRPs configured on the BBP cannot exceed the number of common cells supported by the BBP. The number of common cells supported by each BBP is described in the specifications of the corresponding BBP. For details, see the related BBU technical specifications in <i>3900 & 5900 Series Base Station Product Documentation</i>. If a UBBPg2a serves TRPs for hyper cells and there is at least one 32T32R or 64T64R massive MIMO TRP among these TRPs, a maximum of two massive MIMO TRPs for hyper cells can be configured on the BBP. This is because massive MIMO TRPs consume more processing capability of the BBP.

BBP	Dimension	Configuration Requirement
	Deployment on BBPs	- TRPs for a hyper cell can be deployed on a maximum of two BBPs. Hyper and other cells can be configured on the same BBP. When TRPs for a hyper cell are deployed on two BBPs, the following requirements must be met: - The board names and the settings of the BBP.BBWS, BBP.WM, and BBP.SRT parameters must be consistent between the two BBPs to avoid cell setup failures. - Other TRPs deployed on the two BBPs cannot form another hyper cell or combined cell with the TRPs deployed on a third BBP to avoid cell setup failures. Other cells can be configured on the two BBPs when these BBPs have remaining physical cell processing resources. - It is recommended that one hyper cell be deployed on only one BBP, as shown by scenario 1 in Figure 4-5. This simplifies deployment and prevents deployment failures during cell expansion. If the number of TRPs exceeds the specifications of a BBP, the excess TRPs should be deployed on another BBP, as shown by scenario 2 in Figure 4-5. - When multiple (for example, three) hyper cells are to be deployed, intra-board deployment is recommended, as shown by scenario 1 in Figure 4-6. If inter-board deployment is inevitable, the number of hyper cells each deployed on multiple BBPs must be minimized to prevent cell unavailability caused by insufficient inter-board resources. That is, deploying only one hyper cell on both BBPs is recommended in this case, as shown by scenario 2 in Figure 4-6.

Figure 4-5 Methods of deploying a hyper cell

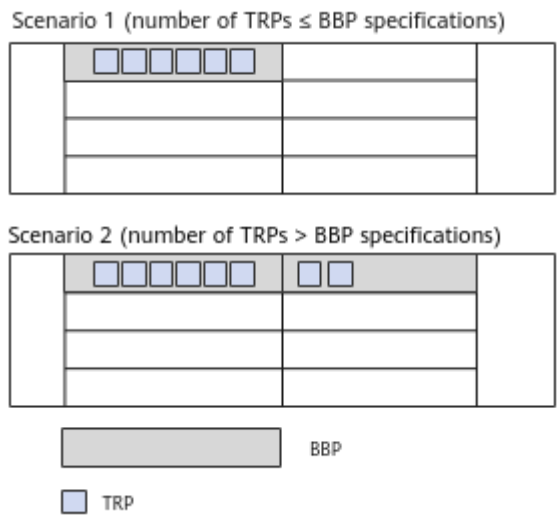
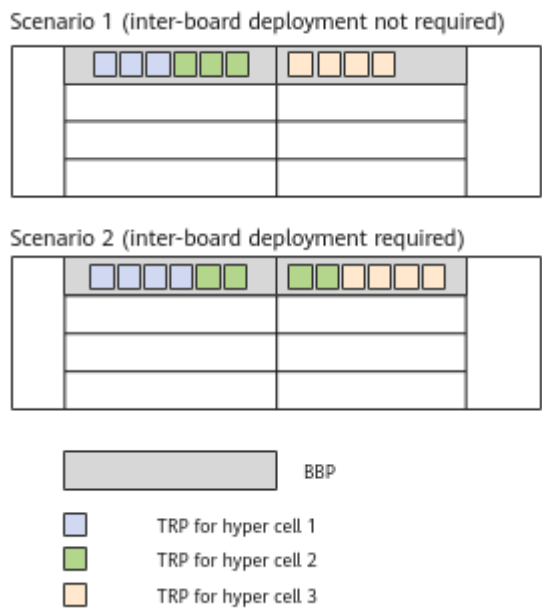


Figure 4-6 Methods of deploying multiple hyper cells



RF Modules

In TDD, RF module requirements are as follows:

- Only NR TDD-capable 1T1R/2T2R/4T4R/8T8R/32T32R/64T64R RF modules support this function. In addition, one of the requirements described in [Table 4-10](#) on the transmit and receive modes of TRPs must be met for the TRPs to form a hyper cell.

Table 4-10 (TDD) Requirements for the transmit and receive modes of TRPs

Scenario	Requirement for Transmit and Receive Modes of TRPs
Low-speed scenario (<i>NRDUCell.HighSpeedFlag</i> set to LOW_SPEED)	• The transmit and receive modes of all TRPs are the same (1T1R/2T2R/4T4R/8T8R). • There are only two transmit and receive modes and the combinations are as follows: – 2T2R and 8T8R – 2T2R and 4T4R – 4T4R and 8T8R
High-speed scenario (<i>NRDUCell.HighSpeedFlag</i> set to HIGH_SPEED)	• The transmit and receive modes of all TRPs are the same (1T1R/2T2R/4T4R/8T8R/32T32R/64T64R). • There are only two transmit and receive modes and the combinations are as follows: – 2T2R and 8T8R – 4T4R and 8T8R – 32T32R and 64T64R

- When 32T32R or 64T64R RF modules are used, eCPRI and CPRI RF modules cannot be configured together.
- In 8T8R or lower scenarios, a maximum of 12 TRPs can be combined to form a hyper cell. In 32T32R or 64T64R scenarios, a maximum of six TRPs can be combined to form a hyper cell. In addition, these TRPs must be configured on BBPs in the same BBU.

In LampSite networking, pRRU565x*AB* and pRRU575x*AB* modules cannot be used together with RF modules beyond these models to form a TRP of a hyper cell. *x* is a digit; *A* and *B* stand for different letters.

The capability to support mixed configuration of TRPs with different transmit and receive modes varies by BBP. For details about the specifications of a BBP, see the corresponding BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation* by choosing **BBU Technical Specifications > Board Technical Specifications > Technical Specifications of Baseband Processing Units > NR Technical Specifications of Baseband Processing Units**. RF modules with different transmit and receive modes must connect to different BBPs if these BBPs do not support mixed configuration.

4.3.4 Networking

- Hyper Cell is applicable in the following scenarios:
 - Linear coverage scenarios, such as high-speed railways and subways
 - LampSite scenarios where the coverage of one TRP does not meet the requirements and multiple TRPs are required to expand the coverage. In this case, Hyper Cell can be used to reduce the network planning complexity.

- Before enabling CA flexible networking, ensure that there is at least two frequencies on the live network. In addition, the cells involved in CA must have overlapping coverage areas and their inter-frequency NR cell relationships must have been configured (using the **NRCellRelation** MO).

4.3.5 Cells

Requirements When a Hyper Cell Is To Be Formed

- SDL cells cannot form a hyper cell.
- The **TRS_RATEMATCH_SW** and **NARROW_BAND_TRS_RATEMATCH_SW** options of the **NRDUCellPdsch.RateMatchSwitch** parameter cannot be both deselected.
- In a hyper cell using the mixed configuration of 2T2R+4T4R or 4T4R+8T8R, the **NRDUCellCsirs.InterTrpCsirsStaggerMode** parameter must be set to **NOT_STAGGER**.
- In TDD:
 - Neither distributed massive MIMO cells nor SUL cells can form a hyper cell.
 - The slot configuration of a hyper cell must be one of the following:
 - 4:1 (with **NRDUCell.SlotAssignment** set to **4_1_DDDSU**)
 - Single-period 8:2 (with **NRDUCell.SlotAssignment** set to **8_2_DDDDDDDSUU**)
 - Dual-period 8:2 (with **NRDUCell.SlotAssignment** set to **8_2_DDDSUUDDDD**)
 - If there are 8T8R or higher TRPs for a hyper cell, it is recommended that the number of SSB beams be the same for these TRPs. That is, the settings of the **NRDUCellTrpBeam.CoverageScenario** parameter (specifying the coverage scenario) and the **NRDUCellTrpBeam.ScenarioBeamAlgoSw** parameter (specifying scenario-based beam algorithm switches) must be consistent among 8T8R or higher TRPs. Otherwise, the access or handover success rate will decrease.
 - If the **NRDUCell.HighSpeedFlag** parameter is set to **HIGH_SPEED** for a cell and the value of an **NRDUCellTrp.TxRxMode** parameter corresponding to the cell is **64T64R**, the **NRDUCellCsirs.FR1MaxCellCsirsPortNum** parameter can only be set to **8PORT** for the cell.

Requirements When Other Functions Are Required

In low-speed scenarios (where the **NRDUCell.HighSpeedFlag** parameter is set to **LOW_SPEED**), if the **SCell_INTER_TRP_HO_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter is selected, that is, inter-TRP handover in CA SCells is enabled, then the **NRDUCellCsirs.CsirsCellResourceNum** parameter must be set to **1_RESOURCE**.

4.3.6 Others

- CA flexible networking has the following requirements for UEs:
UEs must support CA. During initial access, an incoming handover, or an incoming RRC connection reestablishment, a UE reports its CA capabilities to the gNodeB.
- Inter-TRP handover in CA SCells has the following requirements for UEs:
UEs must support at least 16 sets of CSI-RS resources for beam measurement and the reporting of two optimal beams. That is, the values of *maxNumberCSI-RS-Resource* and *maxNumberNonGroupBeamReporting* in the UE-reported capability information must contain **n16** and **n2**, respectively.
- Other functions place no special requirements on UEs. However, if some UEs do not support the configuration of at least six sets of TRS resources or UE-level scrambling codes, problems such as access failures, service drops, or increased BLER may occur. If possible UE compatibility issues such as access failures, frequent service drops, or extremely low traffic volume occur, contact Huawei engineers and engineers from UE manufacturers for handling.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

For details about data preparation, see *Cell Management*. This section only describes the data preparation that is specific to this function.

Table 4-11 and **Table 4-12** describe the parameters used for function activation and optimization, respectively.

Table 4-11 Parameters used for activation

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	NR DU Cell Networking Mode	NRDUCell.NrDuCellNetworkingMode	Set this parameter to HYPER_CELL for a hyper cell and to NORMAL_CELL for a common cell.
Low-frequency TDD	TRS Allocation Strategy	NRDUCellCsirs.TrsAllocationStrategy	It is recommended that this parameter be set to MULTI_TRS .

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Data transmission mode	NRDUCellMultiTrp.DataTr ansMode	Set this parameter to BASIC_MODE or FSD_MODE . It is recommended that this parameter be set to FSD_MODE when inter-TRP interference, weak coverage, or overshoot coverage exists and coverage optimization cannot be performed in the short term.

The parameters in the **NRDUCellTrp** MO must be set for each TRP for the same hyper cell when configuring TRPs for an NR DU cell.

It is recommended that the values of **NRDUCellTrp.NrDuCellTrpld** mod 6 be different for adjacent TRPs to prevent performance issues, such as service drops or decreases in UE throughput caused by TRS interference between TRPs. For example, the **NRDUCellTrp.NrDuCellTrpld** values can be planned for six adjacent TRPs as follows: 15 (mod 6 = 3), 16 (mod 6 = 4), 17 (mod 6 = 5), 18 (mod 6 = 0), 19 (mod 6 = 1), and 20 (mod 6 = 2).

Table 4-12 Parameters used for optimization

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	TRP Handover Delta RSRP	NRDUCellMultiTrp.TrpHo DeltaRsrp	The default value is recommended.

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Inter-TRP Handover RSRP Evaluation Times	NRDUCellMultiTrp.InterTrpHoRsrpEvalTimes	The default value is recommended. When signals fluctuate violently, set this parameter to a larger value to increase the number of evaluation times for an inter-TRP handover, avoiding ping-pong handovers between TRPs. When signals attenuate quickly, set this parameter to a smaller value to reduce the number of evaluation times, ensuring a prompt inter-TRP handover.
Low-frequency TDD	Inter-TRP HO MAC CE Transmission Times	NRDUCellMultiTrp.InterTrpHoMacCeTransTimes	The default value is recommended. For cells with interference or weak coverage, set this parameter to a larger value to increase the number of times MAC CEs are sent to improve the inter-TRP handover reliability.
Low-frequency TDD	TRP HO MAC CE Adaptive Transmission BLER Thld	NRDUCellMultiTrp.TrpHoMacCeAdpTransBlerThld	The default value is recommended.
Low-frequency TDD	TRP HO MAC CE Adaptive Transmission Num Offset	NRDUCellMultiTrp.TrpHoMacCeAdpTransNumOfs	Set this parameter based on the network plan. A larger value of this parameter results in higher overheads for the base station to send TRS MAC CEs, but it causes the reliability of the inter-TRP handovers to rise. It is recommended that this parameter be set to 2 when the uplink and downlink BLERs in a cell are relatively high due to interference.

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Inter-TRP Handover MAC CE Power Offset	NRDUCellMultiTrp.InterTrpHoMacCePwrOfs	The default value is recommended. For cells with interference or weak coverage, set this parameter to a larger value to increase the offset to the power for sending MAC CEs to improve the inter-TRP handover reliability.
Low-frequency TDD	Inter-TRP HO SRS SINR Threshold	NRDUCellMultiTrp.InterTrpHoSrsSinrThld	Set this parameter based on the network plan. If UEs are prone to incorrect TRP handovers due to strong interference on the network, you can modify the parameter value to improve the inter-TRP handover reliability.
Low-frequency TDD	Inter-TRP HO SRS SINR Difference Thld	NRDUCellMultiTrp.InterTrpHoSrsSinrDiffThld	The default value is recommended.
Low-frequency TDD	PUSCH Scramble Optimization Switch	NRDUCellMultiTrp.PuschScrambleOptSw	The default value is recommended. Turn this switch on to enable UEs incompatible with UE-level scrambling codes to access the network.

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Inter-TRP CSI-RS Stagger Mode	NRDUCellCsirs.InterTrpCsirsStaggerMode	For a hyper cell in linear coverage scenarios, if a UE on the network supports the measurement using multiple sets of CSI-RS resources, it is recommended that this parameter be set to MOD2_STAGGER for the base station to stagger resource positions for CSI-RS transmissions from different TRPs. This reduces the number of RRC connection reconfigurations for these UEs. This parameter cannot be set to MOD2_STAGGER in any of the following scenarios: • A hyper cell with the mixed configuration of 2T2R+4T4R or 4T4R+8T8R • A hyper cell with the NRDUCellCsirs.CsirsCellResourceNum parameter set to a value other than 1_RESOURCE • A hyper cell with the NRDUCellCsirs.CsiPeriod parameter set to SLOT10 or SLOT20

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Multi-TRP Algorithm Switch	NRDUCellMultiTrp.MultiTrpAlgoSwitch	- For a macro base station: This parameter applies to continuous coverage. For hyper cells in linear networking, the INTER_TRP_SSB_STAGGER_SWITCH option of this parameter can be selected to enable inter-TRP SSB staggering. When this option is selected, the SSB RSRP decreases and resources occupied by SSBs may increase, causing downlink throughput to decrease. - For a LampSite base station: Inter-TRP SSB staggering is not supported. Even if the INTER_TRP_SSB_STAGGER_SWITCH option of this parameter is selected, this function cannot take effect and performance indicators will be affected. As a result, it is unnecessary to select this option.
Low-frequency TDD	TRS Allocation Strategy	NRDUCellCsirs.TrsAllocationStrategy	The default value is recommended.
Low-frequency TDD	Intra-Hyper-Cell Inter-TRP HO RSRP Policy	NRDUCellMultiTrp.HyperCellTrpHoRsrpPolicy	It is recommended that this parameter be set to DL_EQ_RSRP when the transmit power differs between TRPs.

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Ping-Pong Handover RSRP Thld Offset	NRDUCellMultiTrp.PingPongHoRsrpThldOfs	The default value is recommended.
Low-frequency TDD	Target TRP Overshoot Coverage TA Threshold	NRDUCellMultiTrp.TrpTrpOverCovTaThld	The default value is recommended.
Low-frequency TDD	Target TRP Overshoot Cov HO RSRP Thld Offset	NRDUCellMultiTrp.TrpTrpOverCovHoRsrpThldOfs	The default value is recommended.
Low-frequency TDD	Source TRP Overshoot Coverage TA Threshold	NRDUCellMultiTrp.SrcTrpOverCovTaThld	The default value is recommended.
Low-frequency TDD	Multi-TRP Algorithm Switch	NRDUCellMultiTrp.MultiTrpAlgoSwitch	Set this parameter based on the network plan. When TRP overshoot-coverage optimization is required, the TRP_OVER_COV_HO_OPT_SW option of this parameter needs to be selected.
Low-frequency TDD	Source TRP Overshoot Cov HO RSRP Thld Offset	NRDUCellMultiTrp.SrcTrpOverCovHoRsrpThldOfs	The default value is recommended.
Low-frequency TDD	CA Enhanced Algorithm Switch	NRDUCellCarrMgmt.CaEnhancedAlgoSwitch	Set this parameter based on the network plan. When inter-TRP handover in CA SCells is required, the SCell_INTER_TRP_HO_SW option of this parameter needs to be selected.
Low-frequency TDD	Inter-TRP Handover Delta CSI-RS RSRP Thld	NRDUCellCsirs.InterTrpHoDeltaCsiRsrpThld	The default value is recommended.

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Inter-TRP HO CSI-RS RSRP Evaluation Times	NRDUCellCsirs.InterTrpHoCsiRsrpEvalTimes	The default value is recommended.
Low-frequency TDD	CSI-RS RSRP Filtering Coefficient	NRDUCellCsirs.CsiRsrpFilterCoeff	The default value is recommended.
Low-frequency TDD	CSI-RS Report-based SCC Deactiv Threshold	NRDUCellCsirs.CsirsRptBasedSccDeactivThld	The default value is recommended.
Low-frequency TDD	CA Enhanced Algorithm Switch	NRDUCellCarrMgmt.CaEnhancedAlgoSwitch	Set this parameter based on the network plan. When CA flexible networking is required, the CA_FLEX_NETWORKING_SW option of this parameter needs to be selected.
Low-frequency TDD	Networking Option Opt Sw	gNodeBParam.NetworkingOptionOptSw	In 3CC CA or massive CA co-coverage scenarios, it is recommended that the "CA enhancement in hyper cell co-coverage scenarios or combined cell co-coverage scenarios" function be enabled, that is, the HYPER_CELL_CO_COV_CA_ENH_SW option of this parameter be selected.
Low-frequency TDD	CA Enhanced Algorithm Switch	NRDUCellCarrMgmt.CaEnhancedAlgoSwitch	Set this parameter based on the network plan. When CA flexible data split is required, the CA_FLEX_DATA_SPLIT_SW option of this parameter needs to be selected.

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	High Speed Intrf Random Schedule Sw	NRDUCellAlgoSwitch.HighSpeedIntrfRandomSchSw	It is recommended that high-speed-specific interference randomization-based scheduling be enabled for high-speed hyper cells.
Low-frequency TDD	Multi-TRP Algorithm Switch	NRDUCellMultiTrp.MultiTrpAlgoSwitch	The TRS_CONFIG_OPT_SW option can be selected to improve the access success rate of some UEs. When this option is selected, TRS information is included in the RRC Setup message instead of the RRC Reconfiguration message.
Low-frequency TDD	SRS Period	NRDUCellSrs.SrsPeriod	After a hyper cell is formed, the number of UEs that access the cell increases in most cases. If a fixed SRS period is used, you need to optimize the SRS period configuration based on the maximum number of UEs in the cell to prevent SRS resource allocation failures. Lengthening the SRS period may cause the uplink and downlink cell throughput to decrease. For details about other setting notes for the SRS period, see <i>Channel Management</i> .

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Coverage Scenario	NRDUCellTrpBeam.CoverageScenario	If there are 8T8R or higher TRPs for a hyper cell, it is recommended that the number of SSB beams be the same for these TRPs. That is, the parameter settings of the coverage scenario and scenario-based beam algorithm switches must be consistent among 8T8R or higher TRPs. Otherwise, the access or handover success rate will decrease.
Low-frequency TDD	Scenario-based Beam Algorithm Switch	NRDUCellTrpBeam.ScenarioBeamAlgoSw	
Low-frequency TDD	PUCCH Algorithm Extension Switch	NRDUCellPucch.PucchAlgoExtSwitch	Set this parameter based on the network plan. When inter-TRP RB resource sharing is required, the INTER_TRP_RB_SHARE_POLICY_SW option of this parameter needs to be selected.
Low-frequency TDD	CSI-RS Extend Switch	NRDUCellCsirs.CsRsExtSwitch	Set this parameter based on the network plan. When multi-TRP multi-TRS optimization is required, the MULTI_TRP_MULTI_TRS_OPT_SW option of this parameter needs to be selected.
Low-frequency TDD	PUCCH Algorithm Extension Switch	NRDUCellPucch.PucchAlgoExtSwitch	Set this parameter based on the network plan. When inter-TRP F4 HARQ resource sharing is required, the INTER_TRP_F4_HARQ_SHARE_SW option of this parameter needs to be selected.
Low-frequency TDD	Uplink TRP SRS RSRP Difference Threshold	NRDUCellMultiTrp.UlIntrfTrpSrsRsrpDiffThld	This parameter applies only to the FSD transmission mode. The default value is recommended.

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Downlink TRP SRS RSRP Difference Threshold	NRDUCellMultiTrp.DlIntrfTrpSrsRsrpDiffThld	This parameter applies only to the FSD transmission mode. The default value is recommended.
Low-frequency TDD	CA Enhanced Algorithm Switch	NRDUCellCarrMgmt.CaEnhancedAlgoSwitch	Set this parameter based on the network plan. When adaptive CSI-RS beam sweeping for CA UEs is required, the CA_CSIRS_BM_SWEEP_ADPT_SW option of this parameter needs to be selected.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

During deployment, pay attention to the following:

- During hyper cell deployment or TRP redeployment for a hyper cell, if one of the following situations occurs, modify the networking by referring to the deployment suggestions described in [4.3.3 Hardware](#). If any TRP is unavailable after the modification, run the **STR NRDUCELLREDEPLOY** or **STR BRDBBRESREALLOC** command to redeploy cells on BBPs. Within 15 minutes after either command is executed, if the NR DU cell TRPs are unavailable because redeployment is required due to baseband resource conflicts, cells deployed on the corresponding baseband equipment will be automatically restarted and redeployed. In some conflict scenarios, the redeployment command needs to be executed for two or three times. After that, if any TRP is unavailable and the cell specifications of BBPs are sufficient, contact Huawei engineers for help.
 - Situation 1: When a new cell is set up on a BBU serving a hyper cell, the new cell fails to be activated and the alarm cause is "Redeployment required due to baseband resource conflicts".
 - Situation 2: A TRP for an existing hyper cell is redeployed on another BBP. Cell activation is executed when the cell specifications of the baseband processing unit are sufficient. However, an alarm indicating TRP

unavailability is reported with the alarm cause of "Redeployment required due to baseband resource conflicts".

The **STR NRDUCELLREDEPLOY** command is used to redeploy NR DU cells on the baseband equipment in one click. This is a high-risk MML command and running the command causes some cells served by the BBU to be reestablished. For detailed operations and precautions, see the related MML command reference.

- It is recommended that all cells to be served by the BBU be configured before activating any cell to avoid alarm reporting with the cause of "Redeployment required due to baseband resource conflicts".

Activation Command Examples (TDD)

```
//Adding a TDD hyper cell
ADD NRCELL: NrCellId=0, CellName="CELL1", CellId=1, FrequencyBand=N78, DuplexMode=CELL_TDD;
ADD NRDUCELL: NrDuCellId=0, NrDuCellName="DUCELL1", DuplexMode=CELL_TDD, CellId=1,
PhysicalCellId=1, FrequencyBand=N78, DlnArfcn=634000, UlBandwidth=CELL_BW_100M,
DlBandwidth=CELL_BW_100M, SubcarrierSpacing=30KHZ, SlotAssignment=4_1_DDSU, SlotStructure=SS2,
LogicalRootSequenceIndex=80, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN,
SsbFreqPos=7853, SsbSubcarrierSpacing=30KHZ, NrDuCellNetworkingMode=HYPER_CELL;

//Adding all the TRPs for the hyper cell
ADD NRDUCELLTRP: NrDuCellTrpId=0, TrpType=DEFAULT, NrDuCellId=0, TxRxMode=8T8R,
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=100, CpriCompression=3DOT2_COMPRESSION,
BbResMutualAidSw=ON;
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=0, NrDuCellCoverageId=0, SectorEqmId=0;

ADD NRDUCELLTRP: NrDuCellTrpId=1, TrpType=DEFAULT, NrDuCellId=0, TxRxMode=8T8R,
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=100, CpriCompression=3DOT2_COMPRESSION,
BbResMutualAidSw=ON;
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=1, NrDuCellCoverageId=1, SectorEqmId=1;

//(Optional) Setting the data transmission mode to FSD, which is recommended when TRP interference,
weak coverage, or overshoot coverage exists and coverage optimization cannot be performed in the short
term
MOD NRDUCELLMULTITRP: NrDuCellId=0, DataTransMode=FSD_MODE;

//Setting the TRS allocation strategy for the hyper cell
MOD NRDUCELLCSIRS: NrDuCellId=0, TrsAllocationStrategy=MULTI_TRS;

//Activating the cell
ACT NRCELL: NrCellId=0;
```

Optimization Command Examples

```
//Setting the RSRP difference based on which an inter-TRP handover is triggered in the hyper cell
MOD NRDUCELLMULTITRP: NrDuCellId=0, TrpHoDeltaRsrp=2;

//Setting the number of RSRP evaluations for an inter-TRP handover in the hyper cell
MOD NRDUCELLMULTITRP: NrDuCellId=0, InterTrpHoRsrpEvalTimes=1;

//Setting the number of times MAC CEs are sent, the offset to the number of times MAC CEs are sent, and
the BLER threshold during an inter-TRP handover in the hyper cell
MOD NRDUCELLMULTITRP: NrDuCellId=0, InterTrpHoMacCeTransTimes=2,
TrpHoMacCeAdpTransNumOfs=2, TrpHoMacCeAdpTransBlrThld=25;

//Setting the offset to the power for sending MAC CEs during an inter-TRP handover in the hyper cell
MOD NRDUCELLMULTITRP: NrDuCellId=0, InterTrpHoMacCePwrOfs=0;

//Setting SRS SINR-based evaluation conditions for inter-TRP handovers when the signal quality fluctuates
greatly and the interference is high
MOD NRDUCELLMULTITRP: NrDuCellId=0, InterTrpHoSrsSinrThld=-15, InterTrpHoSrsSinrDiffThld=3;

//Enabling PUSCH scrambling code optimization when there are UEs incompatible with UE-level scrambling
codes on the network
```

```

MOD NRDUCELLMULTITRP: NrDuCellId=0, PuschScrambleOptSw=ON;

//Enabling inter-TRP SSB staggering for the hyper cell served by a macro base station (This function is not
supported by LampSite base stations. Even if the function is enabled, it cannot take effect and performance
indicators will be affected. As a result, it is unnecessary to select the corresponding option in LampSite base
stations.)
MOD NRDUCELLMULTITRP: NrDuCellId=0, MultiTrpAlgoSwitch=INTER_TRP_SSB_STAGGER_SWITCH-1;

//Modifying the CSI-RS configuration policy for the hyper cell
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiPeriod=SLOT40, CsirsCellResourceNum=1_RESOURCE,
InterTrpCsirsStaggerMode=MOD2_STAGGER;

//Enabling inter-TRP handover in CA SCells and setting handover-related parameters (using a low-speed cell
as an example)
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=SCell_INTER_TRP_HO_SW-1;
MOD NRDUCELLCSIRS: NrDuCellId=0, InterTrpHoDeltaCsiRsrpThld=2, InterTrpHoCsiRsrpEvalTimes=1,
CsiRsrpFilterCoeff=20, CsirsRptBasedSccDeactvThld=4;

//Modifying the inter-TRP handover policy when the transmit power differs between TRPs
MOD NRDUCELLMULTITRP: NrDuCellId=0, HyperCellTrpHoRsrpPolicy=DL_EQ_RSRP;

//Setting the offset to the RSRP threshold for an inter-TRP handover to prevent ping-pong handovers
MOD NRDUCELLMULTITRP: NrDuCellId=0, PingPongHoRsrpThldOfs=4;

//Enabling TRP overshoot-coverage handover optimization
MOD NRDUCELLMULTITRP: NrDuCellId=0, MultiTrpAlgoSwitch=TRP_OVER_COV_HO_OPT_SW-1;

//Setting the overshoot-coverage threshold for a target TRP and the offset to the threshold for a handover
to the overshoot-coverage TRP
MOD NRDUCELLMULTITRP: NrDuCellId=0, TrgTrpOverCovTaThld=80;
MOD NRDUCELLMULTITRP: NrDuCellId=0, TrgTrpOverCovHoRsrpThldOfs=0;

//Setting the overshoot-coverage threshold for a source TRP and the offset to the threshold for a handover
from the overshoot-coverage TRP
MOD NRDUCELLMULTITRP: NrDuCellId=0, SrcTrpOverCovTaThld=60;
MOD NRDUCELLMULTITRP: NrDuCellId=0, SrcTrpOverCovHoRsrpThldOfs=0;

//Enabling high-speed-specific interference randomization-based scheduling for the high-speed hyper cell
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, HighSpeedIntrfRandomSchSw=ON;

//Enabling TRS configuration optimization when UE access issues occur
MOD NRDUCELLMULTITRP: NrDuCellId=0, MultiTrpAlgoSwitch=TRS_CONFIG_OPT_SW-1;

//Optimizing the SRS period configuration based on the maximum number of UEs in the hyper cell when a
fixed SRS period is used. The number of UEs that access the cell increases after the combination.
MOD NRDUCELLSRS: NrDuCellId=0, SrsPeriod=SL160;

//Setting the coverage scenario and scenario-based beam algorithm switches in 8T8R or higher scenarios
MOD NRDUCELLTRPBEAM: NrDuCellTrpId=0, CoverageScenario=DEFAULT,
ScenarioBeamAlgoSw=SCENARIO_BEAM_DENSIFY_ALGO_SW-1;

//Enabling CA flexible networking and CA flexible data split
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoSwitch=CA_FLEX_DATA_SPLIT_SW-1&CA_FLEX_NETWORKING_SW-1;
//Enabling the "CA enhancement in hyper cell co-coverage scenarios or combined cell co-coverage
scenarios" function in 3CC CA or massive CA co-coverage scenarios
MOD GNODEBPARAM: NetworkingOptionOptSw=HYPER_CELL_CO_COV_CA_ENH_SW-1;

//Enabling inter-TRP RB resource sharing (only for TDD)
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=INTER_TRP_RB_SHARE_POLICY_SW-1;
//Enabling multi-TRP multi-TRS optimization
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiRsExtSwitch=MULTI_TRP_MULTI_TRS_OPT_SW-1;

//Enabling inter-TRP F4 HARQ resource sharing
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=INTER_TRP_F4_HARQ_SHARE_SW-1;

//(Optional) Setting the uplink TRP SRS RSRP difference threshold and downlink TRP SRS RSRP difference
threshold for the FSD transmission mode
MOD NRDUCELLMULTITRP: NrDuCellId=0, UlIntrfTrpSrsRsrpDiffThld=18, DlIntrfTrpSrsRsrpDiffThld=18;

```

```
//Enabling adaptive CSI-RS beam sweeping for CA UEs  
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=CA_CSIRS_BM_SWEEP_ADPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Deactivating a cell  
DEA NRCELL: NrCellId=0;  
  
//(Optional) Changing the data transmission mode to basic transmission mode if the FSD transmission  
mode is used  
MOD NRDUCELLMULTITRP: NrDuCellId=0, DataTransMode=BASIC_MODE;  
  
//Disabling multi-TRP multi-TRS optimization  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsRsExtSwitch=MULTI_TRP_MULTI_TRS_OPT_SW-0;  
  
//Retaining only one TRP for a common cell and removing the other TRPs  
RMV NRDUCELLCOVERAGE: NrDuCellTrpId=1, NrDuCellCoverageId=1;  
RMV NRDUCELLTRP: NrDuCellTrpId=1;  
  
//Disabling inter-TRP SSB staggering and TRS configuration optimization for the hyper cell  
MOD NRDUCELLMULTITRP: NrDuCellId=0,  
MultiTrpAlgoSwitch=INTER_TRP_SSB_STAGGER_SWITCH-0&TRS_CONFIG_OPT_SW-0;  
  
//Disabling inter-TRP handover in CA SCells  
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=SCELL_INTER_TRP_HO_SW-0;  
  
//Disabling CA flexible networking and CA flexible data split  
MOD NRDUCELLCARRMGMT: NrDuCellId=0,  
CaEnhancedAlgoSwitch=CA_FLEX_DATA_SPLIT_SW-0&CA_FLEX_NETWORKING_SW-0;  
//Disabling the "CA enhancement in hyper cell co-coverage scenarios or combined cell co-coverage  
scenarios" function  
MOD GNODEBPARAM: NetworkingOptionOptSw=HYPER_CELL_CO_COV_CA_ENH_SW-0;  
  
//Disabling inter-TRP RB resource sharing (only for TDD)  
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=INTER_TRP_RB_SHARE_POLICY_SW-0;  
  
//Disabling inter-TRP F4 HARQ resource sharing  
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=INTER_TRP_F4_HARQ_SHARE_SW-0;  
  
//Disabling adaptive CSI-RS beam sweeping for CA UEs  
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=CA_CSIRS_BM_SWEEP_ADPT_SW-0;  
  
//Changing the cell networking mode to common cell  
MOD NRDUCELLCSIRS: NrDuCellId=0, InterTrpCsirsStaggerMode=NOT_STAGGER,  
TrsAllocationStrategy=SINGLE_TRS;  
MOD NRDUCELL: NrDuCellId=0, NrDuCellNetworkingMode=NORMAL_CELL;  
//Disabling TRP overshoot-coverage handover optimization  
MOD NRDUCELLMULTITRP: NrDuCellId=0, MultiTrpAlgoSwitch=TRP_OVER_COV_HO_OPT_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

- Check the cell status.
Run the **DSP NRDUCELL** command to check the value of **NR DU Cell State**. If the output shows that the value is **Normal**, the cell has been successfully set up.
Run the **DSP NRDUCELLTRP** command to check the value of **NR DU Cell TRP State**. If the output shows that the value is **Available**, the TRP has been successfully set up.
- Verify the cell functions.
Observe the counters in [Table 4-13](#). If any counter value is greater than zero, the hyper cell is working properly.

Table 4-13 Counters related to a hyper cell

Counter ID	Counter Name
1911820749	N.HyperCell.TRP.User.RRConn.Avg
1911820750	N.HyperCell.TRP.User.RRConn.Max
1911819893	N.HyperCell.TRP.User.Schedule.DL.Sum
1911819894	N.HyperCell.TRP.User.Schedule.UL.Sum
1911819895	N.HyperCell.TRP.ThpVol.DL.Cell
1911819896	N.HyperCell.TRP.ThpTime.DL.Cell
1911819900	N.HyperCell.TRP.ThpVol.UL.Cell
1911819897	N.HyperCell.TRP.ThpTime.UL.Cell
1911819888	N.HyperCell.TRP.PR.B.DL.Used.Avg
1911819889	N.HyperCell.TRP.PR.B.DL.Avail.Avg
1911819890	N.HyperCell.TRP.PR.B.UL.Avail.Avg
1911819892	N.HyperCell.TRP.PR.B.UL.Used.Avg

- Observe the serving TRP of the UE.
Log in to the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management > NR > User Performance Monitoring > User Common Monitoring**, and view the serving TRP information of the UE.

4.4.3 Network Monitoring

Compare the number of handovers between cells corresponding to the TRPs for hyper cells before the combination with the number of handovers between hyper cells. You can evaluate the decrease in the number of handovers by checking the values of the counters listed in [Table 4-14](#).

Table 4-14 Handover-related counters

Counter ID	Counter Name	Description
1911816906	N.HO.IntragNB.PrepAttIn	Applied to SA networking
1911816918	N.HO.IntragNB.ExecAttIn	
1911816923	N.HO.IntragNB.ExecSuccln	
1911816912	N.HO.IntraFreq.IntragNB.Prep AttOut	
1911816905	N.HO.IntraFreq.IntragNB.Exec AttOut	
1911816907	N.HO.IntraFreq.IntragNB.Exec SuccOut	
1911816750	N.NsaDc.IntraSgNB.PSCell.Ch ange.Att	Applied to NSA networking
1911816751	N.NsaDc.IntraSgNB.PSCell.Ch ange.Succ	

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Channel Management*
- *Downlink Scheduling*
- *Uplink Scheduling*
- *CoMP*
- *Beam Management*
- *Carrier Aggregation*
- *UL and DL Decoupling*
- *Remote Interference Management (Low-Frequency TDD)*
- *Energy Conservation and Emission Reduction*
- *Cell Management*
- *URLLC*
- *Synchronization*
- *MIMO (TDD)*
- *Super Uplink*
- *LTE and NR Power Allocation*
- *Positioning*
- *Extended Cell Range*
- *Mobility Load Balancing in Connected Mode*
- *FWA*
- *Admission Control*
- *3D Networking Experience Improvement*
- *Interference Avoidance*
- *Virtual Grid-based Multi-Frequency Coordination*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *EPC-based NSA Basics*
- *Co-MM (Low-Frequency TDD)*
- *Massive MIMO AHR (Low-Frequency TDD)*

- *Multi-Frequency Smart Selection*
- *Multi-Frequency Smart Aggregation*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *Random Access Control*
- *RedCap*
- *CA Experience Improvement*
- *UE Power Saving*
- *NR Inter-Carrier Dynamic Power Allocation*
- *License Control Item Lists*
- *Uplink Flexible Spectrum Scheduling*
- *High Speed Mobility*
- *Multi-RAT Coordinated Energy Saving*
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *Multi-Operator Sharing*
- *Green Site*
- *Turbo Uplink (Low-Frequency TDD)*
- *5G Networking and Signaling*
- *Standards Compliance*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
- *Network Slicing*
- *Resource Control in NSA Networking*
- *QCI-based Resource Control*
- *BBU Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*